

AI-Enabled Predictive HACCP Using IoT-Based Environmental Monitoring



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Declaration of Originality

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AI Use Declaration

I acknowledge the use of generative artificial intelligence tools, including ChatGPT, during this project. These tools were used to support language improvement, proofreading, document organisation, and the clarification of technical concepts. All AI-generated material was critically reviewed, verified and revised by the author. The research design, implementation, data analysis, interpretation of results and final conclusions are my own work. No confidential or personally identifiable data were submitted to these tools. I accept full responsibility for the accuracy, originality and academic integrity of this dissertation.

Abstract

This research developed an explainable artificial intelligence (AI)- and Internet of Things (IoT)-enabled framework to support predictive Hazard Analysis and Critical Control Point (HACCP) monitoring in cold-storage environments. Three datasets were used: a labelled Fruit Spoilage Dataset for Random Forest and Isolation Forest modelling, the UCI Occupancy Detection Dataset for LSTM temporal benchmarking, and an author-collected refrigerator dataset containing 4,321 IoT observations recorded over 72 hours.

The tuned Random Forest achieved 99.92% accuracy, 100% recall and 99.81% precision in classifying ‘Good’ and ‘Bad’ conditions in the fruit dataset. SHAP analysis identified light, temperature–humidity interaction, humidity and temperature as the most influential predictors. Isolation Forest provided sensitive anomaly detection but produced more false alarms than the supervised model. The 30-observation LSTM achieved 85.40% accuracy and 98.21% recall on the separate UCI occupancy benchmark, demonstrating its ability to model temporal environmental patterns rather than food spoilage directly.

Analysis of the refrigerator dataset identified 17 door-opening episodes and 215 unusual environmental observations. Approximately 91.6% of these anomalies occurred during door opening or within 30 minutes after closure, highlighting the importance of operational context when interpreting alerts. The Raspberry Pi and Azure prototype successfully supported continuous sensing, local indication and cloud telemetry. Overall, the findings demonstrate that supervised prediction, temporal modelling, contextual anomaly detection and explainability can provide complementary early-warning support for predictive HACCP. However, the framework remains a proof of concept and does not replace validated HACCP limits, microbiological testing or professional judgement. The study therefore provides an integrated foundation for future product-specific, longitudinal and microbiologically validated predictive food-safety monitoring.

Keywords: *predictive HACCP, Food Safety, Internet of Things, Random Forest, Isolation Forest, LSTM, SHAP.*

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List of Abbreviations and Acronyms

Abbreviation	Meaning
ANN	Artificial Neural Network
API	Application Programming Interface
AUC	Area Under the Curve
BiLSTM	Bidirectional Long Short-Term Memory
CCP	Critical Control Point
CNN	Convolutional Neural Network
CO ₂	Carbon Dioxide
CSV	Comma-Separated Values
DHT22	Digital Humidity and Temperature Sensor
EDA	Exploratory Data Analysis
FN; FP	False Negative; False Positive
GPIO	General-Purpose Input/Output
HACCP	Hazard Analysis and Critical Control Point
IC	Inter-Integrated Circuit
IF; RF	Isolation Forest; Random Forest
IIoT	Industrial Internet of Things
LCD	Liquid-Crystal Display
LSTM	Long Short-Term Memory
MQTT	Message Queuing Telemetry Transport
PR-AUC	Area Under the Precision–Recall Curve
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SHAP	SHapley Additive exPlanations
TN; TP	True Negative; True Positive
UCI	University of California, Irvine
UPS	Uninterruptible Power Supply
WHO	World Health Organization
XAI	eXplainable Artificial Intelligence

Chapter 1 – Introduction

1.1 Background

Food safety remains a significant challenge for both public health and the food industry. Around 866 million people are estimated to become ill each year after eating contaminated food, with approximately 1.5 million deaths worldwide (World Health Organization [WHO], 2024). This scale makes prevention particularly important. HACCP was developed around this principle, providing a structured way to identify significant hazards and establish controls at critical stages of food production and handling (Codex Alimentarius Commission [CAC], 2022). Yet the effectiveness of the approach still depends heavily on how those controls are monitored. Periodic measurements, fixed limits, manual checks and retrospective verification can leave gaps between observations. If conditions begin to deteriorate during such a gap, the problem may not be recognized until food has already been processed, stored or moved further through the supply chain.

IoT sensing offers a way to reduce this gap. Temperature and humidity, for example, can be recorded continuously rather than checked only at selected times. Operational information can add further meaning to these readings: a rise in refrigerator temperature following a door opening has a different interpretation from a similar rise while the door remains closed. Edge devices and cloud services also make it possible to capture, transmit and analyze these measurements as they occur.

AI extends this idea beyond continuous data logging. Supervised learning can learn relationships between environmental measurements and known spoilage conditions; unsupervised anomaly detection can identify unusual behaviour when labelled outcomes are unavailable; and temporal models can learn how sensor conditions develop over time. Used in this way, predictive HACCP refers to using patterns and emerging changes in monitored conditions to provide earlier warning, rather than simply recording an event or reacting after a fixed threshold has been exceeded.

However, these capabilities do not automatically form an integrated food-safety solution. Existing studies commonly concentrate on individual elements such as IoT sensing, predictive modelling, explainability or deployment, while their integration into an interpretable end-to-end monitoring framework remains less developed. This dissertation addresses that gap by

bringing together environmental sensing, complementary AI approaches, explainability and edge-cloud implementation to investigate conditions associated with elevated food-spoilage risk and abnormal storage behaviour.

1.2 Problem Statement

The central problem is not the availability of environmental sensor data, but how to turn that data into useful early warning. In refrigerated monitoring, a temperature or humidity change may reflect normal operation, such as a door opening, or it may indicate an unusual condition that deserves attention. A model that simply flags the change without this context can generate warnings that are difficult to interpret. Equally, a highly accurate predictive model has limited practical value if its decisions cannot be explained or if it is too demanding to operate within a low-cost monitoring system.

These challenges are complicated by the nature of the available data. Supervised learning requires labelled outcomes, while real IoT streams may contain no confirmed spoilage or anomaly labels. Temporal models can capture how conditions develop over time, but at greater computational cost. The problem is therefore to determine how complementary AI approaches can be used to provide early, interpretable monitoring of environmental conditions associated with spoilage risk, while remaining practical for an IoT and edge-cloud setting.

1.3 Research Aim

This research aims to develop and evaluate an AI-enabled predictive HACCP monitoring framework that integrates environmental IoT sensing with complementary machine-learning approaches to identify conditions associated with elevated food-spoilage risk, supported by explainable AI and implemented within a Raspberry Pi-based edge-cloud environment.

1.4 Research Objectives

To achieve the research aim, the study addresses the following objectives:

1. **To prepare and engineer environmental datasets** containing temperature, humidity and related contextual features for predictive and temporal analysis.
2. **To develop and tune a supervised Random Forest classifier** for identifying environmental conditions associated with elevated food-spoilage risk.
3. **To compare supervised, unsupervised and temporal AI approaches** using Random Forest, Isolation Forest and LSTM to determine their suitability for predictive food-safety monitoring.

4. **To apply SHAP-based explainability** to identify the environmental features that contribute most strongly to model predictions and provide interpretable evidence for the resulting risk classifications.
5. **To implement a Raspberry Pi–Azure IoT prototype** for continuous environmental data acquisition and assess the feasibility of deploying lightweight predictive monitoring at the edge.

1.5 Research Questions

The study is guided by three research questions aligned with the predictive, anomaly-detection and deployment objectives:

- **RQ1:** Can environmental sensor data be used to identify conditions associated with elevated food-spoilage risk?
- **RQ2:** How suitable are supervised, unsupervised and temporal AI approaches for different aspects of predictive food-safety monitoring?
- **RQ3:** How feasible is the deployment of lightweight predictive monitoring on a Raspberry Pi edge device within an IoT environment?

Together, these questions examine whether environmental data can support risk prediction, how complementary AI approaches contribute to monitoring, and whether the resulting approach is practical for resource-constrained edge deployment.

1.6 Scope

This study focuses on using environmental IoT data to support predictive HACCP monitoring. It considers temperature, humidity and door activity, rather than directly measuring microbiological contamination. In the absence of accessible industrial swab data, three complementary datasets are used: a labelled Fruit-spoilage dataset for supervised risk classification, the UCI Occupancy dataset as a temporal benchmark for LSTM sequence modelling, and a 72-hour refrigerator dataset collected through the Raspberry Pi–Azure prototype for real-world environmental monitoring.

The study evaluates Random Forest, Isolation Forest and LSTM, representing supervised, unsupervised and temporal learning, with SHAP used to explain model behaviour. The practical work covers IoT data collection, cloud transmission and the feasibility of lightweight edge deployment. Microbiological testing, industrial-scale or multi-site deployment, and direct detection of chemical or physical hazards are outside the scope. The prototype is therefore evaluated as a proof of concept and does not treat environmental anomalies as confirmed food spoilage or contamination.

1.7 Research Contributions

This research makes the following contributions:

- A unified evidence structure that brings together supervised spoilage-risk classification, unsupervised anomaly detection, temporal modelling and real IoT monitoring to examine predictive HACCP from complementary perspectives.
- A comparative evaluation of Random Forest, Isolation Forest and LSTM, showing the different roles and strengths of supervised, unsupervised and temporal learning rather than presenting them as directly interchangeable models.
- An explainable modelling approach using SHAP, linking predictive results to the environmental factors that influence model decisions.
- A context-aware interpretation of environmental anomalies, using real refrigerator door activity to separate explainable operational disturbances from unusual conditions that may require further attention.
- A practical edge-to-cloud proof of concept, combining Raspberry Pi environmental sensing with Azure telemetry and assessing the feasibility of lightweight edge deployment.

Together, these contributions connect predictive modelling, interpretation and practical IoT monitoring while remaining within the evidence provided by the study.

1.8 Dissertation Structure

This dissertation is organized into seven chapters. Chapter 1 introduces the research problem and sets out the aim, objectives, research questions, scope and main contributions of the study. Chapter 2 reviews relevant work on HACCP, IoT-based environmental monitoring, machine learning and explainable AI before identifying the research gap addressed in this project. Chapter 3 explains the research methodology, including the datasets, preprocessing and feature engineering, model development, evaluation methods and experimental design.

Chapter 4 describes how the proposed system was designed and developed. It covers the system requirements and architecture, Raspberry Pi prototype, data collection and cloud connectivity, as well as the implementation of the Random Forest, Isolation Forest and LSTM models and the approach used to assess edge-deployment feasibility. Chapter 5 presents the experimental results from the AI models and the 72-hour refrigerator deployment, including environmental behaviour, door events and anomaly analysis. Chapter 6 discusses the findings in relation to the research questions, previous studies and the limitations of the work. Finally, Chapter 7 draws together the main conclusions and contributions, reflects on the project and identifies areas for future research.

Chapter 2 – Literature Review

2.1 HACCP and Environmental Monitoring

Hazard Analysis and Critical Control Point (HACCP) is a preventive food safety framework used to identify, assess and control biological, chemical and physical hazards throughout food production. Originally developed during the 1960s to produce safe food for the space programme, HACCP has since been adopted internationally and incorporated into food safety guidance, including the Codex Alimentarius General Principles of Food Hygiene (Codex Alimentarius Commission [CAC], 2022). The framework is based on seven principles covering hazard analysis, identification of Critical Control Points (CCPs), establishment of critical limits, monitoring, corrective actions, verification and documentation (U.S. Food and Drug Administration [FDA], 1997).

Environmental monitoring plays an important role in supporting HACCP by providing information about conditions within food production areas. Surfaces, equipment and food-contact areas can harbour harmful microorganisms, with *Listeria monocytogenes* being a particular concern because of its ability to persist within food-processing environments (Tompkin, 2002). Environmental monitoring therefore uses microbiological sampling alongside measurements of production conditions to assess hygiene and confirm whether preventive controls are working effectively.

Temperature and humidity are particularly relevant because changes in these conditions can influence microbial survival, growth and food quality (Tompkin, 2002). Continuous IoT monitoring can provide real-time environmental data that may support earlier identification of changing conditions (Bouzemrak et al., 2019; Singha et al., 2023). AI, big data and IoT are also being explored as ways of providing earlier and more timely food-safety warning signals (Mu et al., 2024).

Continuous IoT-based monitoring offers a way to observe environmental changes as they happen. When sensor readings are combined with historical data, patterns linked to increased food spoilage risk may be identified and used to provide earlier warnings. This approach could complement existing HACCP controls rather than replace them, forming the central problem this research seeks to address.

2.2 IoT Technologies for Food Safety

The Internet of Things (IoT) has created new opportunities for food safety monitoring by allowing environmental conditions to be measured continuously rather than through periodic manual checks. IoT systems combine sensors, communication networks and data-processing platforms to collect and transmit information from production and storage environments (Atzori et al., 2010). Temperature and humidity are among the most monitored variables because changes in these conditions can affect food quality and create environments that favor spoilage (Bouzembrak et al., 2019).

Recent studies show a shift from basic data collection towards intelligent, real-time monitoring. Ben-Daya et al. (2021) identified considerable potential for IoT in food supply-chain quality management but noted that collecting large volumes of sensor data does not itself support decision-making unless the data can be meaningfully interpreted. More recently, Singha et al. (2023) demonstrated this progression through IntelliStore, combining IoT sensors with machine learning to monitor temperature, humidity, CO₂ and other conditions affecting perishable foods. Such systems show how continuous measurements can provide a richer picture of changing storage conditions than isolated readings (Afreen & Bajwa, 2021). However, IoT monitoring is not without limitations; sensor drift, calibration requirements and variations in data quality can reduce the reliability of low-cost environmental sensors if they are not properly managed (Nalakurthi et al., 2024).

For predictive food safety, the value of IoT lies in the patterns produced over time. Temperature and humidity trends, together with physical events such as door openings, can reveal short-term changes in the monitored environment. A door opening, for example, can disturb the local micro-climate and produce measurable changes in temperature or humidity. Analyzing these patterns provides a basis for investigating whether environmental IoT data can indicate elevated food spoilage risk, directly addressing RQ1. Although wider IoT systems may include

gas, light or other sensors, this project focuses on temperature, humidity and door events to maintain a practical, low-cost monitoring design.

Edge computing further extends this approach by allowing some processing to occur close to the sensors. Tailor et al. (2025), for example, developed a food-quality monitoring system in which temperature, humidity and other sensor measurements were processed through a Raspberry Pi before data was transferred for cloud-based storage and display. Local processing can reduce the amount of data continuously transmitted to the cloud and reduce dependence on network communication for time-sensitive decisions (Shi et al., 2016). Cloud platforms, such as Microsoft Azure, remain useful for longer-term storage and remote access, whereas a resource-constrained Raspberry Pi offers faster local response and greater resilience during temporary connectivity loss. This edge–cloud balance is therefore relevant to RQ3, which examines whether a trained model can operate successfully on a Raspberry Pi for real-time monitoring.

Overall, IoT provides the continuous data required for predictive monitoring, while edge computing offers a practical route for turning those data into timely local decisions. The remaining question is how effectively different AI models can learn from these sensor patterns, which is considered in the following section.

2.3 Machine Learning for Food Safety

The growing availability of sensor and monitoring data has created opportunities to move food safety systems beyond fixed thresholds towards data-driven prediction. Machine learning (ML) can identify relationships within complex datasets that may be difficult to recognize through conventional monitoring alone. Recent studies have applied ML to food quality assessment, spoilage prediction and contaminant detection, showing its potential to support earlier and more informed decision-making (Yang et al., 2025). However, performance depends on the characteristics of the data and the monitoring task, meaning that no single algorithm can be assumed to perform best in every food safety application. This study therefore considers three different approaches: Random Forest, Isolation Forest and Long Short-Term Memory (LSTM).

2.3.1 Random Forest

Random Forest (RF) is a supervised ensemble method that combines predictions from multiple decision trees (Breiman, 2001). Its ability to capture non-linear relationships and interactions between variables makes it suitable for environmental monitoring, where factors such as

temperature and humidity may influence food quality together rather than independently. RF is also relatively robust to noisy data and can provide information about feature importance, making its predictions easier to examine than those of many more complex models. Yi et al. (2024), for example, applied RF to identify potential biomarkers associated with spoilage in refrigerated pork, while recent food-quality research has continued to demonstrate the value of tree-based and ensemble methods (Yang et al., 2025). These characteristics make RF a useful, supervised baseline for determining whether environmental measurements can distinguish between different levels of spoilage risk.

2.3.2 Isolation Forest

Isolation Forest (IF) provides a different perspective because it can identify unusual observations without requiring predefined class labels (Liu et al., 2008). Rather than learning what belongs to a particular class, IF isolates observations that differ from most of the dataset. This is relevant to continuous food monitoring, where abnormal environmental conditions may be relatively rare and labelled examples of every possible event may not be available. Elsaid and Binbusayis (2024) demonstrated the suitability of Isolation Forest for continuously generated IoT sensor data. A key limitation, however, is that an anomaly does not necessarily represent food spoilage. Unusual readings may result from operational changes, sensor errors or genuine environmental variation. Where labelled outcomes are available, comparing detected anomalies against known spoilage conditions can therefore help distinguish meaningful risk patterns from unrelated changes.

2.3.3 Long Short-Term Memory

Long Short-Term Memory (LSTM) is a recurrent neural network designed to learn dependencies within sequential data (Hochreiter & Schmidhuber, 1997). This is particularly relevant to environmental monitoring because how a condition changes over time may be more informative than an individual reading. A sustained rise in temperature, for example, may carry different implications from a brief fluctuation. Yuan et al. (2024) demonstrated the use of LSTM-based architectures for Fruit and vegetable freshness assessment. Environmental monitoring research also supports its suitability for sensor time series, with Guo and Feng (2024) applying LSTM-based modelling to historical temperature and humidity measurements. These studies support the use of LSTM for learning temporal patterns within environmental data.

This capability comes with a practical trade-off. LSTM models generally require greater computational resources and memory than conventional ML approaches and their predictions can be more difficult to interpret. These requirements become particularly important when inference is intended to run on a resource-constrained device such as a Raspberry Pi, where predictive performance must be balanced against processing time and memory usage.

Comparing RF, IF and LSTM therefore provides three complementary perspectives: supervised classification, unsupervised anomaly detection and temporal deep learning. Evaluating them within the same monitoring framework makes it possible to consider not only predictive performance, but also interpretability, computational requirements and suitability for real-time deployment on a resource-constrained edge device.

2.4 Explainable Artificial Intelligence

As machine-learning models become more complex, predictive performance alone may not be sufficient for food safety applications. Practitioners also need to understand why a model has identified a particular condition as risky, especially when an alert may influence monitoring or corrective action. Explainable Artificial Intelligence (XAI) addresses this issue by making model behaviour more transparent and understandable. This transparency is also important for trustworthy AI, as predictions are more useful when the factors behind them can be examined rather than presented simply as unexplained alerts.

Shapley Additive explanations (SHAP) is an XAI method derived from Shapley values in cooperative game theory (Lundberg & Lee, 2017). It estimates how much each input feature contributes to an individual prediction, providing a more principled approach than relying only on basic feature-importance rankings. SHAP has been applied to food-quality prediction (Ahmed et al., 2024) and industrial sensor time-series anomaly detection (Li et al., 2024), supporting its relevance to sensor-based predictive monitoring.

SHAP can provide both global and local explanations. Global explanations reveal which variables generally influence model predictions across a dataset, helping to establish whether factors such as temperature and humidity consistently contribute to predicted spoilage risk. Local explanations can show why a particular real-time alert was generated, making the output more meaningful for staff deciding whether further investigation or action is required.

SHAP can also be adapted to different model types. TreeSHAP provides efficient explanations for tree-based models such as Random Forest, while model-agnostic approaches such as

KernelSHAP can be applied to more complex models. However, generating SHAP values can become computationally expensive for large datasets and complex models, an important consideration for resource-constrained edge deployment. Furthermore, SHAP explains how features influence model predictions but does not establish causal relationships.

In this study, explainability therefore complements conventional performance measures, allowing predictive ability, transparency and practical deployment requirements to be considered together when evaluating the models.

2.5 Review of Recent Studies

Recent research shows a clear shift in food safety and quality monitoring from periodic inspection towards continuous sensing and predictive analysis. Ben-Daya et al. (2021) highlighted the growing role of IoT technologies in food supply-chain quality management, while Wang et al. (2022) reviewed machine-learning applications for monitoring and predicting food-safety risks. Both studies recognized the value of data-driven monitoring but also showed that collecting more data does not necessarily improve decision-making unless the data are reliable and analyzed using suitable methods.

Environmental and sensor data have since become increasingly important. Singha et al. (2023) combined IoT monitoring with machine learning to assess temperature, humidity, CO₂ and other storage conditions affecting perishable foods. Sensor-based approaches were also examined by Manthou et al. (2022), Yuan et al. (2024), Yi et al. (2024) and Doğan et al. (2024) for different spoilage and freshness applications. Although these studies share the aim of detecting food-quality changes earlier, they differ considerably in their sensing methods and experimental settings: some rely on controlled laboratory measurements or specialized sensors, whereas others are closer to continuous environmental monitoring. This variation demonstrates the flexibility of sensor-based prediction, but it also limits direct comparison and raises questions about how well models developed under controlled conditions will perform in real production environments.

Model selection is another recurring issue. Chhetri (2024), Yang et al. (2025) and Balakrishnan et al. (2025) describe applications ranging from Random Forest and Support Vector Machines to neural networks and other deep-learning methods. Yi et al. (2024), for example, used Random Forest to identify variables and potential biomarkers associated with refrigerated pork spoilage, supporting its suitability for modelling complex relationships between measured

features. In contrast, Elsaid and Binbusayyis (2024) demonstrated the usefulness of Isolation Forest for continuously generated IoT data where labelled abnormal events may be limited. LSTM-based methods provide a further alternative by learning how measurements change over time rather than treating each observation independently. Yuan et al. (2024) applied an LSTM-based approach to freshness assessment, while Guo and Feng (2024) demonstrated its ability to learn temporal relationships from temperature and humidity data. These findings suggest that supervised classification, unsupervised anomaly detection and temporal modelling offer different advantages, supporting direct comparison rather than selecting a model solely because it achieved high accuracy in a previous study.

Recent literature also raises concerns about the reliability and interpretation of AI predictions. Nalakurthi et al. (2024) identified calibration, sensor drift and data-quality problems as practical challenges affecting low-cost environmental sensors. Mu et al. (2024) similarly emphasized that reliable data is fundamental to effective AI-supported food-safety monitoring. Explainability has become increasingly important in response to concerns about how predictive decisions are made. Ahmed et al. (2024) used SHAP to interpret food-quality predictions, while Li et al. (2024) applied it to anomaly detection in multivariate industrial sensor time series. These studies show that explanation methods can reveal which measurements influence a prediction. However, such explanations describe model behaviour and should not be treated as evidence that a particular environmental variable caused the observed change in food quality.

Deployment presents another gap between experimental performance and practical monitoring. Tailor et al. (2025) demonstrated a food-quality monitoring architecture in which a Raspberry Pi processed sensor measurements before transferring data to cloud services. This shows that low-cost edge hardware can support real-time monitoring, although prototype validation does not confirm that the same system will remain reliable under the operational demands of a food-manufacturing facility. Recent reviews by Yang et al. (2025) and Balakrishnan et al. (2025) continue to identify challenges involving dataset quality, generalizability, computational requirements and the transfer of experimental AI systems into practical food-production environments.

Overall, existing research provides substantial evidence that IoT sensing, and machine learning can support food-quality and spoilage monitoring. Nevertheless, the literature remains fragmented, with many studies concentrating on sensing, prediction, explainability or deployment separately. Few compare supervised, unsupervised and temporal models within

the same environmental monitoring framework while also considering interpretability and deployment on low-cost edge hardware. This creates a clear basis for examining whether environmental sensor data can predict elevated spoilage risk, which model provides the most appropriate balance of performance and interpretability, and whether the resulting model can operate effectively in a practical real-time monitoring system.

Table 1 - Comparison of Selected AI & IoT Approaches for Predictive Food-Safety & HACCP Monitoring

Study	Application / Dataset	Model / Technology	Main Finding	Limitation Relevant to This Project
Ben-Daya et al. (2021)	Food supply-chain quality management	IoT literature review	IoT enables continuous, data-rich monitoring of food quality across supply chains.	Limited translation of collected sensor data into practical decision support.
Wang et al. (2022)	Food-safety monitoring literature	RF, SVM, ANN and other ML approaches	Machine learning shows strong potential for food-risk monitoring and prediction.	Heterogeneous datasets and evaluation methods restrict direct comparison.
Manthou et al. (2022)	Ready-to-eat leafy vegetables	Spectroscopy with PLSR and SVR	Sensor and ML combinations can predict microbiological spoilage.	The best sensor–model combination varied by food type.

Singha et al. (2023)	Ten Fruits and vegetables under monitored storage	IoT with SVM, decision tree, AdaBoost and gradient boosting	Real-time environmental monitoring supported quality assessment; SVM achieved the strongest reported classification result.	Controlled data and limited external validation reduce generalizability.
Yuan et al. (2024)	Fruit and vegetable freshness	CNN–BiLSTM deep learning	Temporal deep-learning methods can support freshness assessment.	The application is not a direct match for low-cost temperature and humidity sensing.
Yi et al. (2024)	Refrigerated pork spoilage	Random Forest	RF identified variables and potential biomarkers associated with spoilage.	The dataset is product-specific and may not transfer directly to other foods.
Nalakurthi et al. (2024)	Low-cost environmental sensing	Review of sensor calibration methods	Calibration is essential for obtaining reliable measurements	Sensor drift and changing environmental conditions can reduce accuracy.

			from low-cost sensors.	
Doğan et al. (2024)	Fish spoilage monitoring	Colorimetric sensor with embedded ML	The system enabled rapid, on-device spoilage classification.	It depends on specialized sensing materials rather than basic environmental sensors.
Ahmed et al. (2024)	Sweet potato quality assessment	Hyperspectral imaging, machine learning and SHAP	SHAP helped identify the features influencing food-quality predictions.	Explanations describe model behaviour but do not establish causality.
Li et al. (2024)	Industrial multivariate sensor time series	Anomaly detection with SHAP	The study demonstrated explainable anomaly detection for continuous sensor streams.	The industrial dataset was not food specific.
Elsaid & Binbusayyis (2024)	Continuously generated IIoT sensor data	Isolation Forest	IF was suitable for identifying uncommon patterns in streaming sensor data.	A detected anomaly does not necessarily represent spoilage or food-safety risk.
Guo & Feng (2024)	Environmental temperature	LSTM-based modelling	LSTM captured	Training and inference require more

	and humidity time series		temporal dependencies in environmental measurements.	computational resources than lighter models.
Taylor et al. (2025)	Food-quality monitoring prototype	Raspberry Pi, ESP32 and cloud services	A low-cost edge-cloud architecture supported real-time sensor monitoring.	Validation remained at prototype scale rather than full industrial deployment.
Yang et al. (2025)	Food safety, quality and nutrition literature	Machine-learning review	Ensemble and deep-learning methods show substantial predictive potential.	Dataset quality, comparability and generalizability remain recurring challenges.
Balakrishnan et al. (2025)	AI-enabled food-safety sensing	Smart sensors, spectroscopy and AI review	AI can improve rapid detection and predictive food-safety monitoring.	High-dimensional sensing increases calibration and model-complexity demands.

2.6 Critical Analysis of Existing Research

The literature demonstrates the potential of IoT and machine learning for food-quality monitoring, but it also reveals several limitations that influenced the design of this project. Many studies use small, product-specific or laboratory-controlled datasets, making it difficult to determine whether reported performance will transfer to other environments (Balakrishnan et al., 2025). Access to industrial food-production data is also restricted by commercial

sensitivity and the limited availability of matched environmental and microbiological records. Consequently, this project uses two complementary public datasets and one author-collected IoT refrigerator dataset rather than relying on a single source. The public datasets support labelled food-risk classification and temporal benchmarking, while the refrigerator dataset provides real-world environmental, door-event and anomaly-monitoring evidence from the implemented prototype.

The Multi-Parameter Fruit Spoilage Dataset was selected as the primary dataset because it contains measurements directly associated with changes in food condition and provides labelled spoilage outcomes for supervised evaluation. However, its product-specific nature limits wider generalization. The UCI Occupancy Detection Dataset is therefore used as a benchmark because it contains time-stamped environmental measurements, including temperature and humidity, collected from a real indoor sensing environment (Candanedo & Feldheim, 2016). Although occupancy is not equivalent to food spoilage, the dataset enables the monitoring framework to be tested in a different IoT context. It is not used as microbiological evidence, but as a way of assessing temporal sequence modelling on a separate, time-ordered environmental IoT dataset across contrasting sensor datasets.

The choice of Random Forest, Isolation Forest and LSTM also responds to limitations in previous research. Studies frequently report the performance of one model under a particular set of conditions, making it difficult to judge whether its success results from the algorithm or the dataset. Comparing the three models introduces supervised classification, unsupervised anomaly detection and temporal deep learning within the same framework. Random Forest provides an interpretable and computationally efficient baseline (Breiman, 2001); Isolation Forest addresses situations where labelled risk events are limited (Liu et al., 2008); and LSTM examines whether patterns developing over time offer additional predictive value (Hochreiter & Schmidhuber, 1997). Their differences also allow predictive performance to be considered alongside computational cost.

SHAP was included because accuracy alone provides limited support for food-safety decisions. Explaining how temperature, humidity and other variables influence individual predictions can make alerts easier to assess and may reveal whether a model has learned plausible relationships (Lundberg & Lee, 2017). Nevertheless, these explanations are treated as descriptions of model behaviour rather than evidence of causation.

Finally, deployment on a Raspberry Pi addresses the limited practical validation found in much of the literature. Testing inference time, memory use and local alert generation moves the project beyond offline modelling and examines whether the selected model can operate on affordable, resource-constrained hardware (Shi et al., 2016). Together, these decisions create a balanced evaluation of predictive ability, transparency, generalizability and practical deployment.

2.7 Research Gap

The reviewed literature demonstrates that IoT sensors and machine-learning models can support food-quality assessment, spoilage prediction and environmental monitoring. However, existing research remains fragmented. Many studies use specialized sensors, controlled laboratory conditions or product-specific datasets, while others focus mainly on predictive accuracy without examining explainability or practical deployment (Wang et al., 2022; Chhetri, 2024; Balakrishnan et al., 2025). As a result, there is limited evidence showing whether accessible environmental measurements, such as temperature and humidity, can provide meaningful early indications of elevated spoilage risk across different sensing contexts. This limitation leads to the first research question:

RQ1: Can environmental sensor data be used to identify conditions associated with elevated food-spoilage risk?

The literature also provides no consistent agreement about the most suitable modelling approach. Supervised models perform well when labelled outcomes are available, unsupervised methods can identify unusual conditions where labels are scarce, and temporal models can learn patterns that develop over time. However, these approaches are rarely compared within the same monitoring framework using consistent evaluation measures (Wang et al., 2022; Yang et al., 2025). This leads to the second research question:

RQ2: How suitable are supervised, unsupervised and temporal AI approaches for different aspects of predictive food-safety monitoring?

A further gap concerns the transition from offline experiments to practical monitoring systems. Many studies evaluate models on desktop or cloud-based platforms without considering whether they can operate efficiently on affordable edge hardware. Predictive performance alone does not establish whether a model can deliver timely results under the processing and

memory constraints of a Raspberry Pi (Tailor et al., 2025; Balakrishnan et al., 2025). This leads to the third research question:

RQ3: How feasible is the deployment of lightweight predictive monitoring on a Raspberry Pi edge device within an IoT environment?

Together, these questions address the linked challenges of predictive value, model selection and practical deployment.

The project therefore contributes an integrated framework that combines three complementary data sources, two public datasets and one author-collected 72-hour IoT refrigerator dataset, with supervised classification, unsupervised anomaly detection, temporal modelling, SHAP-based explanation and Raspberry Pi–Azure implementation.

Chapter 3 - Research Methodology

3.1 Research Methodology

This study adopted an experimental, quantitative methodology to develop and evaluate an AI- and IoT-enabled predictive HACCP framework. The work combined environmental data collection, preprocessing, feature engineering, machine-learning model development, validation, testing and performance evaluation.

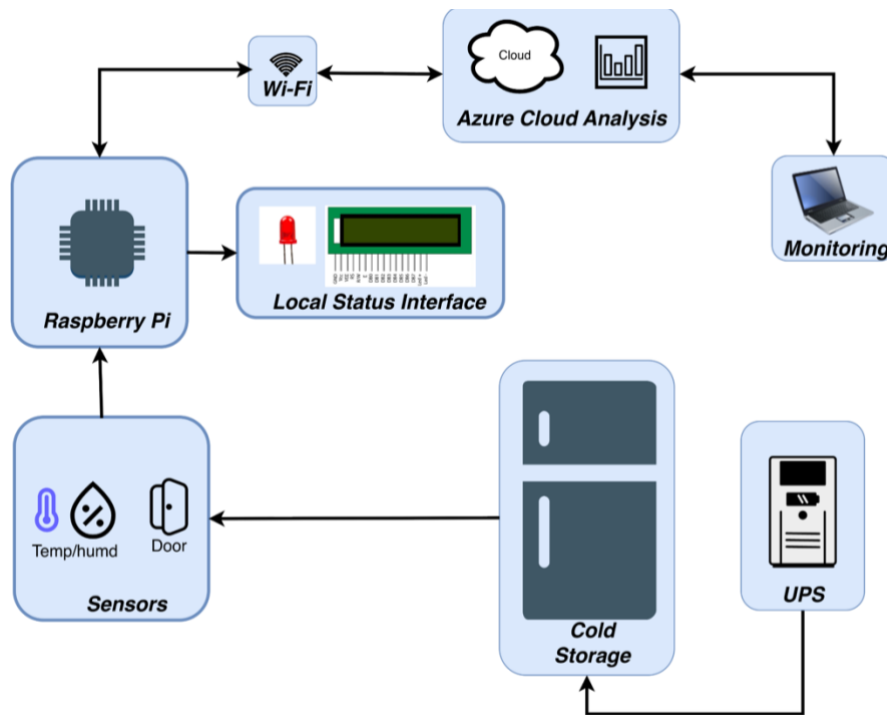
Three datasets were used for different parts of the study. The Fruit Spoilage Dataset supported supervised spoilage-risk classification using Random Forest and unsupervised anomaly

detection using Isolation Forest. The UCI Occupancy Detection Dataset provided a time-ordered benchmark for LSTM sequence modelling. The third source was the prototype refrigerator dataset, which contained temperature, humidity and door-status data collected during a 72-hour deployment using the Raspberry Pi monitoring system.

The Raspberry Pi recorded the refrigerator measurements locally and transmitted telemetry through Wi-Fi to Azure IoT Hub. This dataset supported the analysis of real environmental conditions, sensor-location differences and door-related disturbances. Azure was used for telemetry collection rather than model training or cloud-based AI analysis. Model development and evaluation were completed offline.

3.2 Overall Research Framework

The study adopted two complementary workflows: an IoT monitoring workflow and a machine-learning workflow. The IoT workflow, presented in Figure 1, consists of



environmental and door sensors installed in the cold-storage unit, a Raspberry Pi for data acquisition and local logging, Wi-Fi communication, Azure IoT Hub for cloud telemetry, and a local monitoring interface. A UPS supports continuity during power interruptions. Together, these components enable continuous collection and remote monitoring of temperature, humidity and door-state measurements.

Figure 1 - IoT architecture for cold-storage environmental monitoring and Azure telemetry

The machine-learning workflow is shown in Figure 2 and uses three datasets. The Fruit Spoilage Dataset follows a tabular pathway involving preprocessing, feature engineering, stratified data partitioning, and Random Forest and Isolation Forest modelling. The UCI Occupancy Detection Dataset follows a temporal pathway involving preprocessing, sequence preparation, chronological partitioning and LSTM development. The prototype refrigerator dataset provides time-ordered temperature, humidity and door-state measurements collected by the Raspberry Pi. It supports context-aware anomaly analysis under real household operating conditions.

The models are evaluated on data not used during training. The selected measures include accuracy, precision, recall, F1-score, PR-AUC and confusion matrices, with measures applied according to the task and available labels. SHAP and feature-importance analysis are used to interpret model behaviour and assess its relevance to the proposed predictive HACCP framework.

3.3 Dataset Selection and Justification

Three datasets were selected to address the food-related, temporal and real-world monitoring objectives of the research.

The Multi-Parameter Fruit Spoilage Dataset was selected as the primary food-related dataset because it contains environmental measurements associated with labelled good and bad Fruit conditions. Its variables include Fruit type, temperature, humidity, light and CO₂. This made it suitable for supervised spoilage-risk classification using Random Forest and unsupervised anomaly detection using Isolation Forest.

The raw dataset contained 10,995 observations. After inconsistent class labels were corrected and 2,880 duplicate records were removed, 8,115 unique observations remained. The main limitations are the absence of timestamps and microbiological measurements. Its target is therefore interpreted as a Fruit-condition or spoilage-risk label rather than direct evidence of microbial contamination.

The UCI Occupancy Detection Dataset was selected as a temporal benchmark. It contains 20,560 timestamped environmental observations recorded at approximately one-minute intervals. Its variables include temperature, humidity, light, CO₂ and humidity ratio. The

dataset was used to develop and evaluate the LSTM model using chronological training, validation and test partitions. Because occupancy is not a food-safety outcome, this dataset was used only to demonstrate temporal environmental modelling and detection-delay assessment.

The prototype refrigerator dataset was collected during the 72-hour Raspberry Pi deployment. It contains 4,321 timestamped observations from two temperature and humidity sensors, together with the refrigerator door state. The dataset was used for real-world environmental monitoring and context-aware anomaly analysis. It has no ground-truth spoilage or anomaly labels, so its findings are interpreted as unusual environmental patterns rather than confirmed food-safety events.

Table 2 - Comparison of the selected machine-learning datasets

Criterion	Multi-Parameter Fruit Spoilage Dataset	UCI Occupancy Detection Dataset	Prototype refrigerator dataset
Role in the study	Primary food-related dataset	Temporal benchmark dataset	Real-world operational dataset
Raw observations	10,995	20,560	4,321
Data structure	Tabular and non-temporal	Time-ordered sensor data	Time-ordered dual-sensor and door data
Main variables	Fruit type, temperature, humidity, light and CO ₂	Temperature, humidity, light, CO ₂ and humidity ratio	Temperature ×2, humidity ×2 and door state
Target variable	Good/Bad Fruit condition	Occupied/Unoccupied	None; unlabeled
Timestamp available	No	Yes	Yes
Models applied	Random Forest and Isolation Forest	LSTM	Context-aware Isolation Forest

Criterion	Multi-Parameter Fruit Spoilage Dataset	UCI Occupancy Detection Dataset	Prototype refrigerator dataset
Partitioning strategy	Stratified training, validation and testing	Chronological training, validation and testing	Not applicable; unsupervised operational analysis
Main purpose	Spoilage-risk classification and anomaly detection	Temporal modelling and detection-delay evaluation	Environmental and door-event monitoring
Main limitation	No timestamps or microbiological ground truth	Not a food-spoilage or food-safety dataset	No anomaly or spoilage ground truth

Together, the three datasets provide complementary forms of evidence. The Fruit Spoilage Dataset supports food-related risk modelling, the UCI dataset tests temporal sequence learning, and the prototype refrigerator dataset demonstrates monitoring and anomaly analysis using data collected from the implemented system. Their results are interpreted separately because the datasets have different variables, targets and analytical purposes.

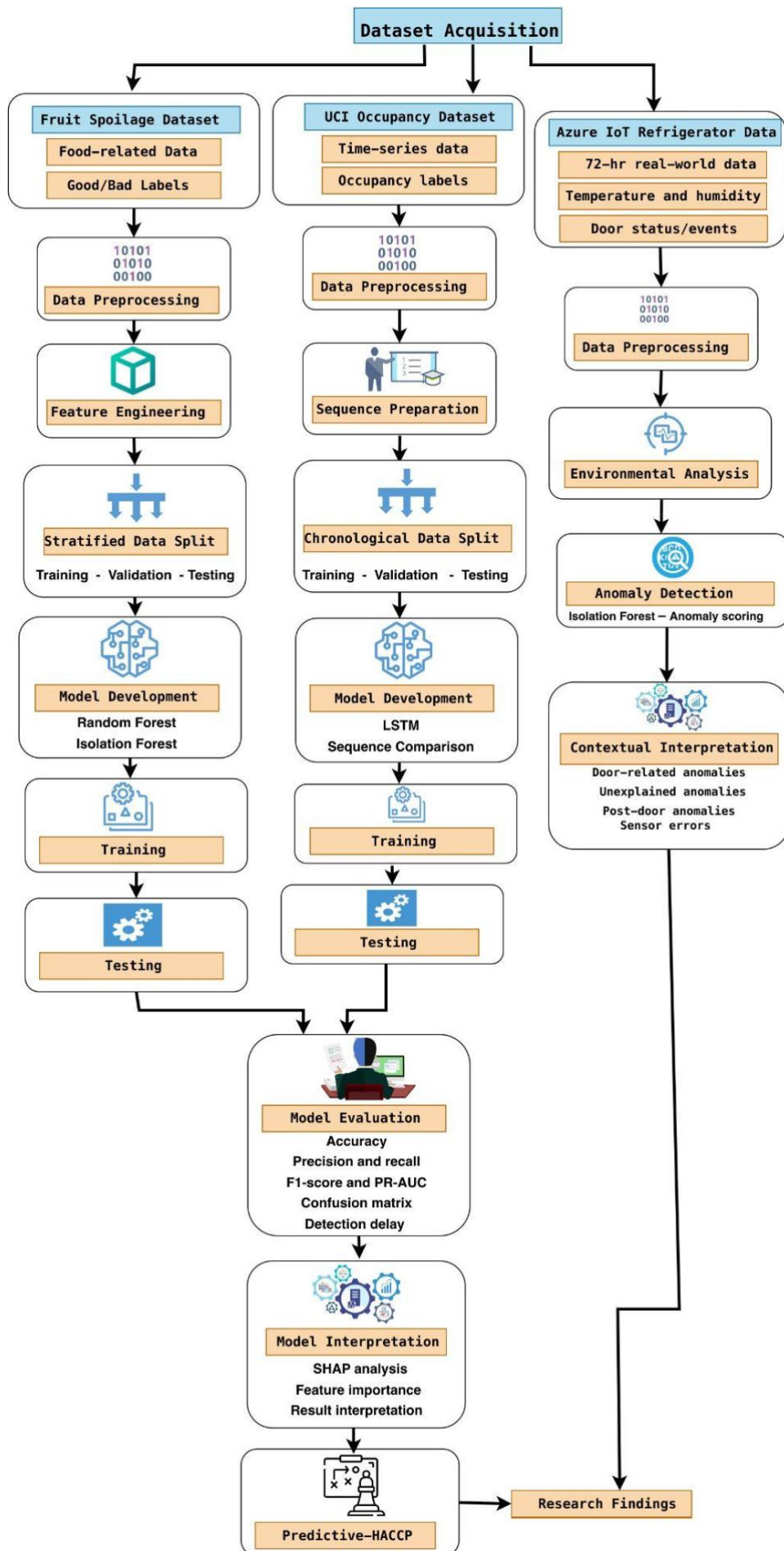


Figure 2 - Machine-learning workflow for the Fruit Spoilage, UCI Occupancy and Azure IoT refrigerator datasets

3.4 Data Pre-processing

Pre-processing was performed separately for the three analytical datasets because each supported a different part of the research design. The Multi-Parameter Fruit Spoilage Dataset supplied the labelled food-related observations used for supervised classification and label-assisted evaluation of anomaly detection. The UCI Occupancy Detection Dataset supplied an external time-series benchmark for the LSTM. The prototype refrigerator dataset supplied real operational measurements for IoT system assessment, environmental monitoring, door-event analysis and context-aware unsupervised anomaly detection. Occupancy and environmental outliers were not interpreted as food-spoilage or microbiological labels.

All source files were retained unchanged. Reproducible Python scripts wrote cleaned and engineered versions to separate processed-data directories, thereby preserving the raw evidence and allowing each transformation to be traced.

3.4.1 Multi-Parameter Fruit Spoilage Dataset

The raw Fruit dataset (Marius, et al., 2025) contained 10,995 observations and six variables: Fruit type, temperature, relative humidity, light intensity, CO₂ concentration and spoilage class. Column names were standardized. The inconsistent label BAD was consolidated with Bad, after which the outcome was encoded as Good = 0 and Bad = 1.

No values were missing, but 2,880 rows were exact duplicates. These were removed before data partitioning so that identical records could not appear in both model-development and test sets. The cleaned dataset contained 8,115 unique observations: 4,569 Good cases (56.3%) and 3,546 Bad cases (43.7%). Broad plausibility checks were applied to identify recording errors rather than impose HACCP critical limits. The observed ranges were 21-27°C for temperature, 71-95% for humidity, 4.22-268.45 lux for light and 20-478 ppm for CO₂; no additional observations were removed.

3.4.2 UCI Occupancy Detection Dataset

The UCI benchmark (Candanedo L., 2016) comprised three files containing 20,560 observations: 8,143 training records, 2,665 observations in the first test file and 9,752 in the second. Timestamps were parsed, observations were sorted within each file, numeric columns were converted explicitly, and the occupancy target was converted to an integer. No missing

values or exact duplicates were found. The original file boundaries were preserved rather than concatenating and randomly redistributing the observations. This protected temporal order and prevented neighboring time points from crossing partitions.

3.4.3 Prototype Refrigerator Dataset

The Prototype Refrigerator Dataset contained 4,321 records collected over 72.00 hours from two temperature-humidity sensors and a door-state sensor. Timestamp parsing, chronological sorting, schema validation, duplicate checking and sampling-interval auditing were performed first. The median interval was 60 seconds, the maximum was 73 seconds, and no interval exceeded 90 seconds. On the assumed one-minute schedule, all 4,321 expected records were present. This completeness measure describes the Prototype Refrigerator Dataset; it is not a record-level Raspberry Pi-to-cloud delivery rate because the matching local Pi log was unavailable.

Sensor 1 contained three error records, with temperature and humidity missing in the same observations. Sensor 2 contained no errors or missing values. Raw missingness and error indicators were retained. For feature computation only, short internal gaps in Sensor 1 were time-interpolated with a two-observation limit. Door status was normalized to OPEN or CLOSED, and agreement between *'door_event'* and *'door_status'* was verified.

3.4.4 Prevention of Data Leakage

All transformations requiring fitted parameters were learned from model-development data only. The encoders and numeric transformations for supervised learning were contained within Scikit-learn pipelines. The Fruit isolation forest standardizer was fitted to the training partition. The LSTM scaler was fitted only to the official UCI training file and reused unchanged on the other files. The prototype Isolation Forest used complete, sensor-error-free observations for fitting; door state and error flags were deliberately withheld from the feature matrix and introduced only after detection for contextual interpretation.

3.5 Feature Engineering

Feature engineering was guided by the physical meaning of environmental monitoring and by the input requirements of tabular, temporal and edge-deployment models.

3.5.1 Fruit Spoilage Features

Two interaction variables were created:

$$\text{Temperature-humidity interaction} = \text{Temperature} \times \text{Humidity}$$

$$\text{CO}_2\text{-light interaction} = \text{CO}_2 \times \text{Light}$$

The first represents combined thermal and moisture conditions, while the second exposes a possible joint relationship between CO₂ and illumination. These terms do not imply causality; they give the algorithms explicit representations of relationships that might otherwise require multiple tree splits. Fruit type was treated as nominal and one-hot encoded during fitting.

3.5.2 UCI Temporal Features

Temperature, humidity, light and CO₂ were lagged by 1, 5 and 10 observations. Ten-observation rolling means, standard deviations, minima and maxima were calculated, together with first differences, hour of day and day of week. Rows made incomplete by lagging were removed only after feature generation. The engineered table contained 41 variables. The LSTM used a focused set of 16 current, difference, rolling and time-of-day inputs. Explicit lags were omitted from the final recurrent input because each sliding sequence already supplied consecutive historical observations. Sequences of 10 and 30 observations were evaluated.

3.5.3 Refrigerator Environmental and Operational Features

The refrigerator analysis retained separate sensor values so that spatial differences were not erased by immediate averaging. It also calculated mean temperature, mean humidity, Sensor 1-minus-Sensor 2 differences, one-step rates of change, 1-, 5- and 15-minute deltas, and rolling means, standard deviations and ranges over 5, 15 and 30 minutes. These variables represented current level, local trend, volatility and sensor disagreement.

Consecutive OPEN records were grouped into distinct door episodes. For each episode, the procedure calculated start and closure time, estimated duration, a robust pre-event baseline from recent closed-door observations, maximum temperature and humidity response, time to temperature increase and post-closure recovery. Recovery required both sensors to remain within robust baseline tolerances for five consecutive observations and was searched for a maximum of 120 minutes or until the next opening. Door-open status, the first 30 minutes after closure and sensor-error flags were retained as contextual variables rather than model inputs.

3.5.4 Raspberry Pi-compatible Feature Subset

The reduced edge model used variables reproducible from DHT22 measurements, reed-switch events and the Raspberry Pi clock: temperature, humidity, their interaction, rolling means and standard deviations, first differences, door-open count and hour of day. The Fruit dataset was

not time-stamped and contained no door records, so several offline deployment features were positional approximations. The reduced model was therefore a software and feature-feasibility experiment rather than a fully validated longitudinal refrigerator-risk model.

3.6 Model Development

The modelling strategy combined supervised classification, unsupervised detection, temporal learning and reduced edge inference. A random seed of 42 was used for conventional machine-learning experiments.

3.6.1 Random Forest and Supervised Baselines

Random Forest was the primary classifier for the labelled Fruit task. A pipeline one-hot encoded Fruit type transformed numeric variables and fitted the classifier. Balanced class weighting reduced bias towards good observations. Hyperparameters were examined over 100, 200 and 300 trees; unrestricted, 10 and 20 maximum depths; leaf sizes of 1, 2 and 4; sqrt and log2 feature sampling; and balanced weighting alternatives. Recall for the Bad class was the optimization measure because false negatives represent potentially risky conditions classified as normal. A fixed predefined validation fold ensured that tuning used the training and validation partitions only. The selected model used 100 trees, unrestricted depth, sqrt feature sampling, minimum leaf size 1 and balanced class weights. Majority-class and balanced Logistic Regression models provided reference baselines.

3.6.2 Fruit Isolation Forest

The Fruit Isolation Forest examined detection when abnormal labels were unavailable during fitting. It was trained only on good training observations. Candidate contamination settings ranged from 0.05 to 0.48, with alternative tree counts, subsample sizes and feature fractions. Validation recall selected the operating configuration, but precision and false-positive rate were reported because contamination directly influences alert frequency. Predictions were mapped to anomalous/normal and compared with Bad/Good labels only after fitting.

3.6.3 LSTM Temporal Benchmark

The LSTM contained recurrent layers of 64 and 32 units, each followed by 0.30 dropout and batch normalization, then a 32-unit rectified-linear layer and sigmoid output. Adam optimization and binary cross-entropy loss were used with class weighting. Sequence lengths of 10 and 30 were trained for up to 30 epochs with batch size 128. Early stopping monitored validation loss, restored the best weights and worked with learning-rate reduction on a plateau.

A 0.50 threshold converted probabilities to binary occupancy predictions. This experiment tested temporal modelling capability, not food spoilage.

3.6.4 Context-aware Refrigerator Isolation Forest

An additional Isolation Forest was fitted to the unlabeled refrigerator measurements using only environmental and temporal variables. The model used 300 trees, standardized features, contamination 0.05 and random seed 42. Because the file contains no ground-truth anomaly outcome, contamination was treated as an analytical operating choice rather than an observed hazard prevalence. Sensitivity runs were completed at 0.01, 0.025, 0.05, 0.075 and 0.10.

After scoring, each flagged observation was labelled as occurring during a door opening, within 30 minutes after closure, during a sensor error, or during unexplained closed-door operation. Adjacent flags were aggregated into anomaly periods. This separation between detection and interpretation prevented the model from simply learning the known door-state label and allowed unusual environmental behavior to be examined against real operational context.

3.6.5 Reduced Edge Model

A reduced Random Forest with 200 trees, maximum depth 15 and minimum leaf size 2, was trained using the Raspberry Pi-compatible features. The fitted scaler and model were exported together as a *joblib* artefact so that inference applied the same transformation as model development.

3.7 Model Validation Strategy

The cleaned Fruit data was divided using a stratified 70:15:15 train-validation-test design, producing 5,680 training, 1,217 validation and 1,218 held-out test observations. Hyperparameters were selected on the validation partition, after which the test partition was evaluated once. The same seed and partitions were used for supervised baselines. Random Forest and label-assisted evaluation of Isolation Forest. Predefined Split functioned as a transparent fixed train-validation fold rather than ordinary random k-fold cross-validation.

For the UCI benchmark, it has been split in to training, validation and testing for evaluation. Sliding windows stayed within their source file. The first test file precedes the training file chronologically, so it is an official independent validation set but not strict forward-chaining validation. Both sequence lengths were reported; selecting the stronger final-test result is acknowledged as a limitation.

The Refrigerator dataset had no normal/anomaly labels and therefore could not support supervised accuracy, recall or cross-validation. Validation instead combined methodological checks: reproducible fixed-seed fitting, contamination sensitivity analysis, exclusion of sensor-error observations from fitting, contextual comparison with door events, event-level aggregation and investigation of unexplained closed-door periods. This is a plausibility and operational-context validation strategy, not ground-truth confirmation of spoilage.

3.8 Evaluation Strategy

For labelled classification, the positive class was 'bad' or 'occupied'. Accuracy, precision, recall and F1-score were calculated from true-positive, true-negative, false-positive and false-negative counts. Recall was prioritized for the Fruit-risk task because it measures the proportion of bad cases detected, whereas precision quantified the credibility of alerts. PR-AUC assessed positive-class ranking, ROC-AUC described discrimination across thresholds, and confusion matrices exposed the nature of errors. Results were compared directly only when models addressed the same dataset and target.

The SHAP tree explainer was applied to the fitted random forest. Global mean absolute SHAP values, a summary plot, dependence plots and local waterfall examples were produced. SHAP values explain the fitted model's output; they do not demonstrate that a measurement caused spoilage.

For the refrigerator deployment, evaluation covered four distinct dimensions. Data-system evidence comprised coverage, sampling regularity, completeness within the refrigerator file and sensor-error frequency. Environmental evidence comprised distributions, trends, dual-sensor correlation and systematic location differences. Operational evidence comprised door-episode count, duration, response magnitude and recovery. Unsupervised evidence comprised anomaly score, number and proportion of flags, sensitivity to contamination, proportion explained by door/error context and the timing of unexplained periods. Environmental flags were never interpreted as confirmed contamination or HACCP failure.

McNemar's test was not calculated because paired observation-level predictions were not retained and the test cannot be reconstructed from aggregate confusion matrices. Food-spoilage detection delay was also unsupported because the Fruit data lack timestamps and verified hazard-onset times. Refrigerator door response describes environmental recovery, not spoilage detection delay.

Chapter 4 - System Design and Development

4.1 Functional Requirements

The functional requirements define the operations that the proposed prototype and analytical workflow were expected to perform. Table 3 summarizes these requirements from environmental sensing through to data analysis.

Table 3 - Functional requirements of the proposed system

ID	Functional requirement	Purpose
FR1	The system shall measure temperature and humidity using two environmental sensors.	Monitor conditions at two locations within the refrigerator.
FR2	The system shall detect and record whether the refrigerator door is open or closed.	Provide operational context for changes in temperature and humidity.
FR3	The system shall collect and timestamp sensor readings at approximately one-minute intervals.	Produce continuous time-series data for monitoring and analysis.
FR4	The Raspberry Pi shall store sensor readings locally.	Maintain a local record and provide a backup of collected telemetry.
FR5	The system shall display current sensor readings and door status on the LCD.	Allow direct checking of prototype operation during deployment.
FR6	The Raspberry Pi shall transmit collected telemetry through Wi-Fi to Azure IoT Hub for cloud storage.	Support remote collection and later analysis of IoT data.
FR7	The analytical workflow shall support RF, IF and LSTM modelling.	Evaluate supervised, unsupervised and temporal approaches.
FR8	The analytical workflow shall provide SHAP-based explanations for supported predictive models.	Improve interpretation of model decisions.
FR9	The system shall support preparation of a reduced model for Raspberry Pi deployment.	Assess the feasibility of lightweight edge inference.

Note. *FR1-FR6 concerns the data-acquisition prototype, while FR7-FR9 concerns the analytical workflow.*

4.2 Non-functional Requirements

The non-functional requirements define the expected quality attributes and practical constraints of the prototype and analytical workflow.

Table 4 - Non-functional requirements of the proposed system

ID	Non-functional requirement	Design consideration
NFR1 - Reliability	The system should support continuous environmental monitoring with minimal interruption.	Local logging is retained alongside cloud transmission.
NFR2 - Sampling consistency	Measurements should be collected at approximately one-minute intervals.	Provides sufficiently regular temporal data for subsequent analysis.
NFR3 - Data integrity	Stored records should retain timestamps, sensor values, door information and sensor-error indicators.	Enables data-quality checking and identification of failed sensor readings.
NFR4 - Efficiency	The monitoring system should operate within the limited computing resources of a Raspberry Pi.	Supports the intended low-resource edge setting.
NFR5 - Low cost	The prototype should use accessible, relatively inexpensive sensing and computing hardware.	Supports practical proof-of-concept development.
NFR6 - Interpretability	AI predictions should be supported by understandable feature-level explanations where applicable.	SHAP is used to examine model behaviour.
NFR7 - Reproducibility	Data collection, preprocessing and modelling steps should be documented and repeatable.	Supports verification of the experimental results.
NFR8 - Security	Device credentials and cloud-access information should not be exposed within datasets or reported outputs.	Protects access to the Azure IoT environment.

4.3 System Architecture

The system was designed as an edge-to-cloud architecture that links environmental sensing, local processing, cloud telemetry and offline AI analysis. Figure 3 presents the overall data flow, while Figure 4 shows the physical prototype used during data collection.

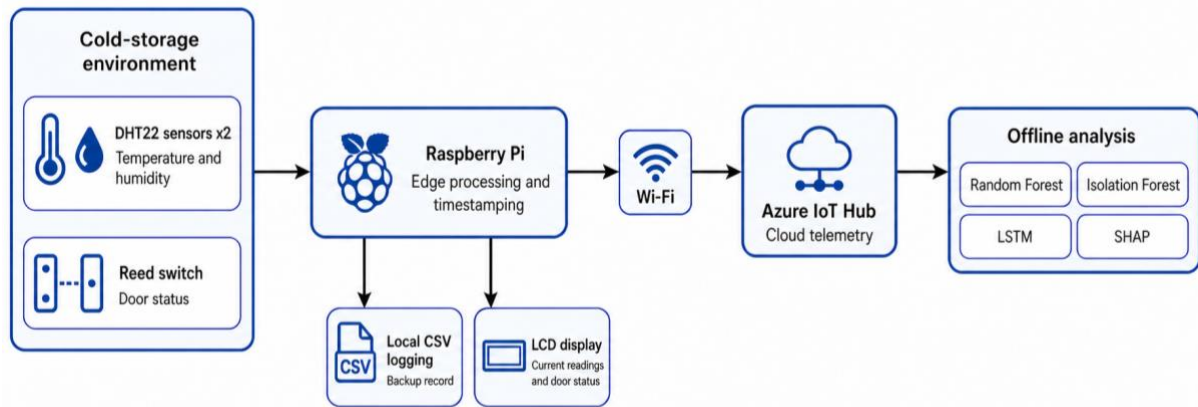


Figure 3 - Edge-to-cloud architecture of the predictive HACCP monitoring system.

Within the cold-storage environment, two DHT22 sensors measure temperature and humidity at different positions. A reed switch records whether the refrigerator door is open or closed. These components satisfy FR1 and FR2 by collecting environmental measurements together with door activity.

The sensors are connected to a Raspberry Pi, which acts as the main processing device. It reads and timestamps the measurements at approximately one-minute intervals, addressing FR3 and NFR2. The readings are stored locally in a CSV file, as required by FR4. Local storage also provides a backup if cloud communication is interrupted and therefore supports the reliability requirement in NFR1. An LCD displays the current sensor readings and door status, meeting FR5 and allowing the prototype to be checked during operation.

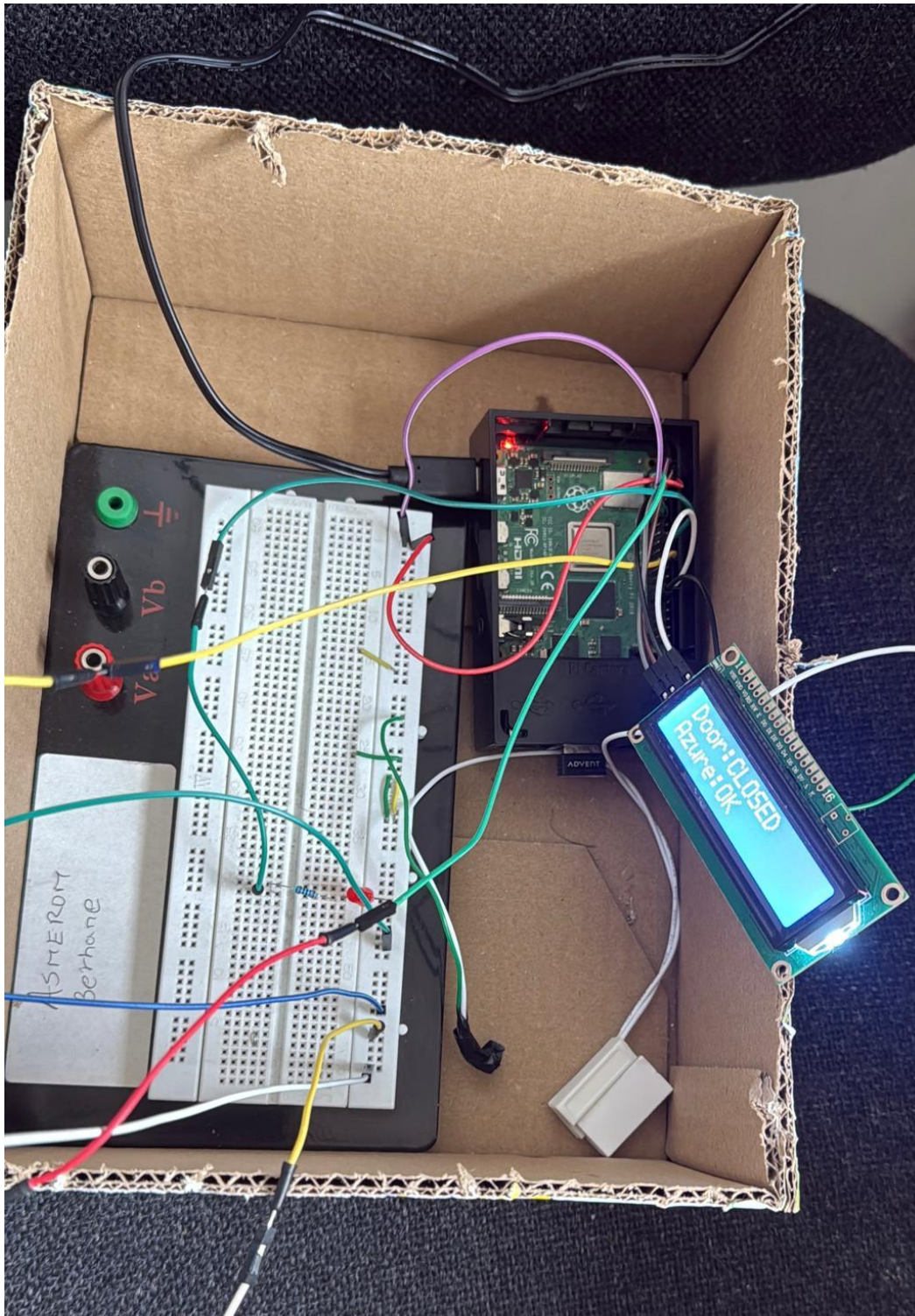


Figure 4 - Physical implementation of the Raspberry Pi monitoring prototype, including the breadboard circuit, LCD and door sensor

The Raspberry Pi sends the collected telemetry through Wi-Fi to Azure IoT Hub, satisfying FR6. This connects the physical prototype to the wider data-processing workflow. Device credentials are kept separate from the collected datasets and reported results in line with NFR8.

After collection, the data are exported for preprocessing and offline analysis. Random Forest is used for supervised risk classification, Isolation Forest for anomaly detection, and LSTM for temporal sequence modelling. SHAP provides feature-level explanations where applicable. This analytical stage addresses FR7, FR8 and NFR6. A reduced model is also prepared to assess the feasibility of future Raspberry Pi deployment under FR9. However, the main model training and evaluation are performed offline; the prototype is not presented as having completed live AI inference.

4.4 Raspberry Pi Prototype

The physical prototype shown in Figure 4 was built around a Raspberry Pi 4. While Section 4.3 described the overall movement of data through the system, this section explains how the monitoring device was assembled and operated during testing.

Two DHT22 sensors were connected to the Raspberry Pi to measure temperature and relative humidity. One sensor was placed near the refrigerator door to capture disturbances caused by opening, while the other was positioned towards the back to monitor more stable conditions. This arrangement allowed spatial differences to be examined rather than assuming that conditions were uniform throughout the refrigerator. A magnetic reed switch was used to detect whether the door was open or closed, providing operational context for short-term environmental changes.

A breadboard and jumper wires connected the sensors, reed switch, LCD and LED to the Raspberry Pi. The LCD displayed the latest readings and door status, allowing the prototype to be checked directly. The LED was used only as a simple development-time indicator during testing and was not treated as a separate functional requirement. The components were arranged inside a temporary cardboard enclosure to organize the wiring and reduce accidental movement. This construction was suitable for a proof-of-concept study but was not intended to represent an industrial product.

The Raspberry Pi ran the data-collection program, timestamped each observation and stored it locally in a CSV file. The intended sampling interval was approximately one minute. During the 72-hour deployment, the median interval was verified as 60 seconds, with a maximum interval of 73 seconds. Local logging reduced the risk of losing all collected data if network communication was interrupted.

The program also updated the LCD and transmitted telemetry through Wi-Fi to Azure IoT Hub. The device credential used for this connection was kept separate from the logged dataset. Azure acted as the cloud communication endpoint, while preprocessing and AI evaluation were completed offline.

Overall, the prototype provided a practical implementation of FR1–FR6. Its local backup, low-cost components and modest processing requirements also supported NFR1, NFR4, NFR5 and NFR8.

4.5 Data Collection Prototype

The prototype was deployed in a single domestic refrigerator for 72 hours to determine whether the complete system could collect continuous environmental data under realistic household conditions. The deployment provided a practical proxy for cold-storage monitoring rather than an industrial HACCP trial.

Temperature and humidity were monitored using two DHT22 sensors. One was positioned near the refrigerator door, while the other was placed towards the back. This arrangement captured conditions at an exposed location and a more stable internal location instead of relying on a single measurement point.

The refrigerator remained in normal household use throughout the deployment. Most door openings therefore occurred during everyday use. A small number of openings were also introduced deliberately to observe whether the sensors and reed switch captured the resulting environmental changes. This provided known disturbances without presenting the deployment as a fully controlled experiment. The reed switch recorded the door state alongside each environmental observation, allowing changes in temperature and humidity to be interpreted in relation to door activity.

The deployment also recorded one exceptional door-open episode lasting approximately 257 minutes. This event was not treated as representative of normal refrigerator use, but it provided an opportunity to examine the system's response to a prolonged disturbance. Its effect and limitations are discussed further in Chapters 5 and 6.

Measurements were scheduled at approximately one-minute intervals. Each observation contained a timestamp, readings from both sensors and the corresponding door state. The structure of the records also allowed sensor errors to be identified during data-quality checks. Data was stored locally on the Raspberry Pi and transmitted through Wi-Fi to Azure IoT Hub. Local storage provided a backup, while Azure supplied the cloud telemetry path.

The resulting dataset supported later evaluation of environmental stability, differences between sensor locations, door-related disturbances and unusual behaviour. The achieved sampling consistency, number of observations, door-opening episodes and environmental responses are reported in Chapter 5.

4.6 Random Forest Implementation

The Random Forest classifier was implemented using the cleaned Fruit spoilage dataset to distinguish between good and bad Fruit conditions. A preprocessing and modelling pipeline was constructed to encode categorical variables and process the numerical features consistently. Hyperparameters, including the number of trees, tree depth, minimum split size

and feature-selection strategy, were tuned using the validation set. The final model was then evaluated once on the independent test set and saved for subsequent SHAP interpretation.

4.7 Isolation Forest Implementation

Isolation Forest was implemented as an unsupervised method for detecting unusual environmental conditions in the Fruit Spoilage Dataset. The model was trained using only observations labelled Good, while the class labels were excluded from model fitting. The number of trees, contamination level, sample size and feature proportion were tuned using validation data. The resulting anomaly predictions were subsequently compared with the known ‘good’ and ‘bad’ labels for evaluation. This approach demonstrated anomaly detection without requiring labelled abnormal observations during training.

A separate Isolation Forest was applied to the unlabeled Azure IoT refrigerator dataset. It used environmental and temporal features derived from the two temperature and humidity sensors. Door status and sensor-error flags were excluded from model fitting and were used after anomaly scoring to interpret each detected observation. The model used 300 trees and a contamination setting of 0.05, with sensitivity analysis performed across alternative contamination values. Because the dataset contained no ground-truth anomaly or spoilage labels, the resulting flags were interpreted as unusual environmental observations rather than confirmed food-safety events.

4.8 LSTM Implementation

The LSTM model was implemented using the time-ordered UCI Occupancy Detection Dataset as a temporal benchmark. Numerical features were scaled using parameters learned from the training data and sliding sequences of 10 and 30 observations were prepared. A two-layer LSTM network incorporating dropout and batch normalization was trained using chronological training and validation partitions. Early stopping and learning-rate reduction were applied to limit overfitting. The selected model was finally evaluated on chronologically separate test data. Occupancy prediction was treated as temporal environmental modelling rather than food-spoilage prediction.

4.9 Edge AI Deployment

A reduced Random Forest model was prepared to assess the feasibility of edge AI deployment on the Raspberry Pi. The model was limited to 200 trees, a maximum tree depth of 15 and a

minimum of two observations per leaf. It also used a Pi-compatible feature subset restricted to information that could be reproduced from the DHT22 sensors, reed switch and Raspberry Pi clock.

The reduced model and its preprocessing objects were exported together as a *joblib* artifact. This allowed new observations to be processed using the same feature structure and scaling rules applied during offline modelling. The exported artifact provided a practical basis for later integration with the Raspberry Pi data-collection program.

The reduced model was evaluated offline before export. It achieved a recall of 0.9737, compared with 1.0000 for the full Random Forest model. This small reduction in recall showed the trade-off between predictive performance and the simpler model configuration required for a resource-constrained edge device. Random Forest was selected instead of the LSTM for this assessment because tree inference mainly involves a sequence of comparisons, whereas an LSTM requires recurrent weight calculations and greater memory and processing capacity.

In a complete live deployment, the Raspberry Pi would collect temperature, humidity and door-status data, create the required input features and pass them to the exported model. The resulting prediction could provide a local indication of elevated environmental risk. Predictions could then be displayed locally or transmitted with the original telemetry to Azure IoT Hub.

Continuous live risk prediction was not implemented during the 72-hour refrigerator deployment. The Raspberry Pi completed data acquisition, local logging, display output and cloud transmission, while model training, evaluation and inference testing remained offline. FR9 was therefore addressed through measured deployment feasibility rather than a completed real-time prediction service. Future work should integrate the exported model into the live monitoring program and evaluate prediction latency, reliability and long-term behaviour on the Raspberry Pi.

Chapter 5 - Experimental Results

This chapter reports the food-related classification results, the temporal benchmark and the real-world Azure refrigerator deployment. The Fruit dataset supplies labelled spoilage-risk evidence; UCI supplies a temporal IoT benchmark; and prototype supplies unlabeled environmental and operational evidence. These tasks are complementary but not interchangeable.

5.1 Exploratory Data Analysis

5.1.1 Fruit and UCI datasets

After duplicate removal, the Fruit dataset contained 8,115 observations: 4,569 Good and 3,546 Bad. The descriptive statistics are summarized in Table 5. Bad observations had higher mean temperature (24.35 compared with 23.42°C), humidity (94.95% compared with 91.96%) and light (38.20 compared with 10.97 lux). Target correlations were strongest for humidity ($r = 0.459$), temperature ($r = 0.363$) and light ($r = 0.312$); CO₂ had little linear association ($r = 0.013$). The UCI files contained 20,560 time-ordered observations, of which 23.1% were occupied. Light showed the strongest association with occupancy ($r = 0.915$), followed by temperature and CO₂, supporting its use as a separate temporal benchmark.

Table 5 - Descriptive statistics for the cleaned Fruit dataset

Statistic	Temperature (°C)	Humidity (%)	Light (lux)	CO ₂ (ppm)
Mean	23.824	93.266	22.870	325.690
Standard deviation	1.274	3.236	43.314	59.752
Minimum	21.000	71.000	4.221	20.000
Median	24.000	95.000	12.836	331.000
Maximum	27.000	95.000	268.448	478.000

5.1.2 Prototype refrigerator dataset

The refrigerator dataset contained 4,321 observations spanning 72.00 hours, with no duplicates and no interval above 90 seconds. Sensor 1 had three error/missing records and Sensor 2 had none. This demonstrates sustained collection and cloud-side storage with very low data loss within the observed refrigerator dataset. However, exact edge-to-cloud delivery reliability cannot be claimed without matching each record to a local Raspberry Pi log.

The sensors were strongly correlated but not interchangeable. Sensor 1 averaged 5.51°C and Sensor 2 averaged 7.37°C; their correlation was 0.985 and the mean Sensor 1-minus-Sensor 2 difference was -1.85°C. During stable closed-door periods the difference was -1.92°C, consistent with a persistent spatial effect. The separate sensor traces were therefore retained.

Door operation produced a marked environmental contrast. Table 6 summarises the sensor data, while Figure 6 shows the 72-hour temperature profiles. Across OPEN observations, mean temperatures were 17.93°C for Sensor 1 and 18.49°C for Sensor 2, compared with 4.29 and 6.21°C during closed periods outside the 30-minute recovery context. Mean humidity during the post-door window was 67.55% and 64.57%, higher than the corresponding stable closed period means of 56.17% and 52.62%. These descriptive results establish that the door variable represents a physically meaningful operational disturbance rather than an arbitrary metadata field.

Table 6 - Refrigerator data-quality and dual-sensor summary

Measure	Result
Monitoring records	4,321
Coverage	72.00 hours
Median / maximum interval	60 / 73 seconds
Sensor 1 / Sensor 2 errors	3 / 0
Mean Sensor 1 / Sensor 2 temperature	5.51 / 7.37°C
Temperature correlation	0.985
Mean Sensor 1-minus-Sensor 2 difference	-1.85°C

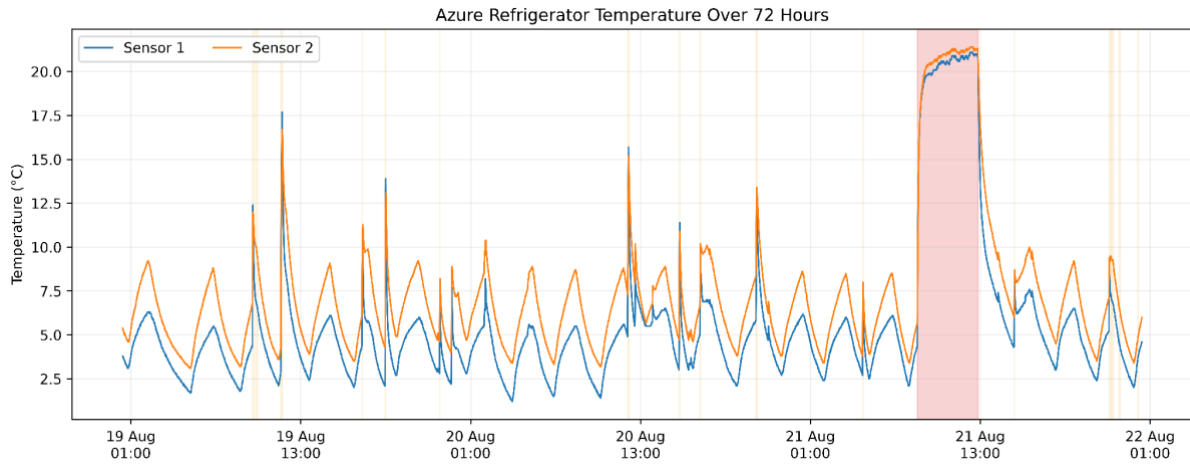


Figure 5 - Temperature measurements from two refrigerator locations across the 72-hour Azure deployment

5.2 Random Forest Results

The tuned random forest was evaluated on 1,218 held-out Fruit observations. The test metrics are reported in Table 7, with the confusion matrix shown in Figure 7. It correctly classified all 532 Bad cases and 685 of 686 Good cases. Recall was 1.0000, precision 0.9981, F1-score 0.9991 and accuracy 0.9992; PR-AUC and ROC-AUC were both 1.0000. Validation performance was similarly high, and default and tuned configurations produced the same recorded validation metrics. The result therefore reflects strong separability in the dataset rather than a large gain from tuning.

Table 7 - Random Forest test performance

Recall	Precision	F1-score	PR-AUC	ROC-AUC	Accuracy
1.0000	0.9981	0.9991	1.0000	1.0000	0.9992

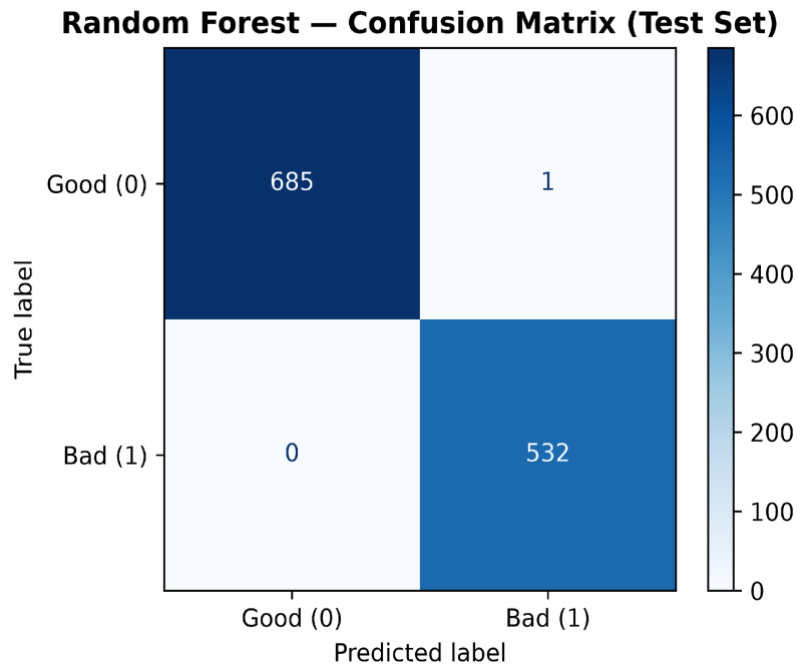


Figure 6 - Confusion matrix for the tuned Random Forest on the held-out Fruit test set

Impurity importance ranked the temperature-humidity interaction first (0.2271), followed by humidity (0.2063), light (0.1885), CO₂-light interaction (0.1313), temperature (0.0986) and CO₂ (0.0633). This feature-importance ranking is shown in Figure 8. The model therefore relied mainly on environmental measurements and their interactions, although impurity importance does not establish direction or causality.

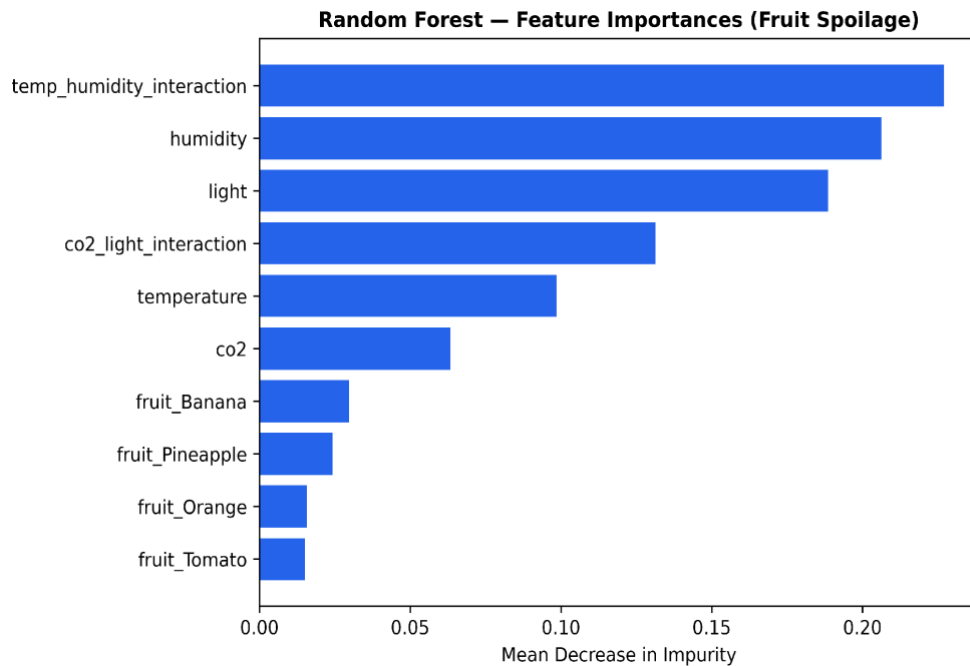


Figure 7 - Impurity-based feature importance for the tuned Random Forest

5.3 Isolation Forest Results

5.3.1 Label-assisted Fruit Evaluation

The Fruit Isolation Forest detected 525 of 532 Bad cases, giving recall of 0.9868. Table 8 reports the evaluation metrics, while Figure 9 shows the anomaly-score distribution. It also flagged 330 of 686 Good cases. Precision was consequently 0.6140, F1-score 0.7570, accuracy 0.7233, ROC-AUC 0.8175 and PR-AUC 0.6849. The false-positive rate of 0.4810 demonstrates the operational cost of configuring an unsupervised detector for very high sensitivity.

Table 8 - Fruit Isolation Forest test performance

Recall	Precision	F1-score	PR-AUC	ROC-AUC	Accuracy	False-positive rate
0.9868	0.6140	0.7570	0.6849	0.8175	0.7233	0.4810

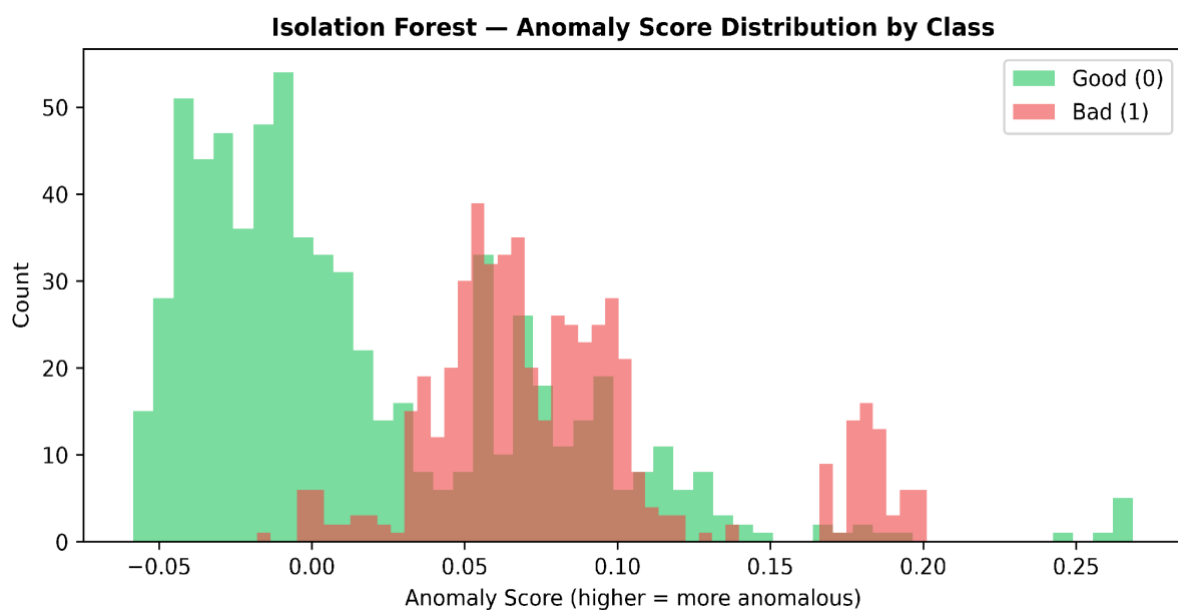


Figure 8 - Fruit Isolation Forest anomaly-score distribution by recorded class

5.3.2 Refrigerator contextual anomaly detection

At contamination 0.05, the refrigerator Isolation Forest model flagged 215 observations. Table 9 summarises the anomaly contexts, which are visualised in Figure 9 and Figure 10. Sixty occurred during OPEN periods and 137 within 30 minutes after closure. Thus, 197 of 215 flags (91.6%) were associated with known door context; no model flag coincided with the three sensor-error records. Eighteen observations (8.4%) remained unexplained during recorded closed-door operation. Sensitivity analysis showed that the number of flags rose from 43 at

contamination 0.01 to 429 at 0.10, confirming that contamination is an analytical control rather than a measured hazard rate.

Table 9 - Context of refrigerator environmental anomaly observations

Context	Observations	Percentage of anomaly observations
During door opening	60	27.9%
Within 30 minutes after closure	137	63.7%
Sensor error	0	0.0%
Unexplained closed-door operation	18	8.4%
Total	<u>215</u>	<u>100.00%</u>

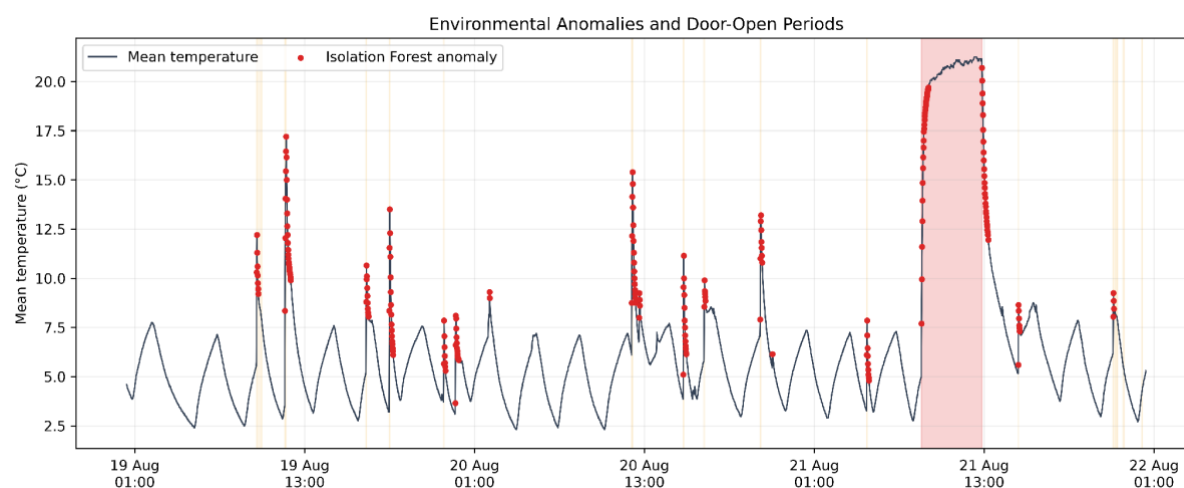


Figure 9 - Refrigerator environmental anomaly observations shown against mean temperature and door-open periods

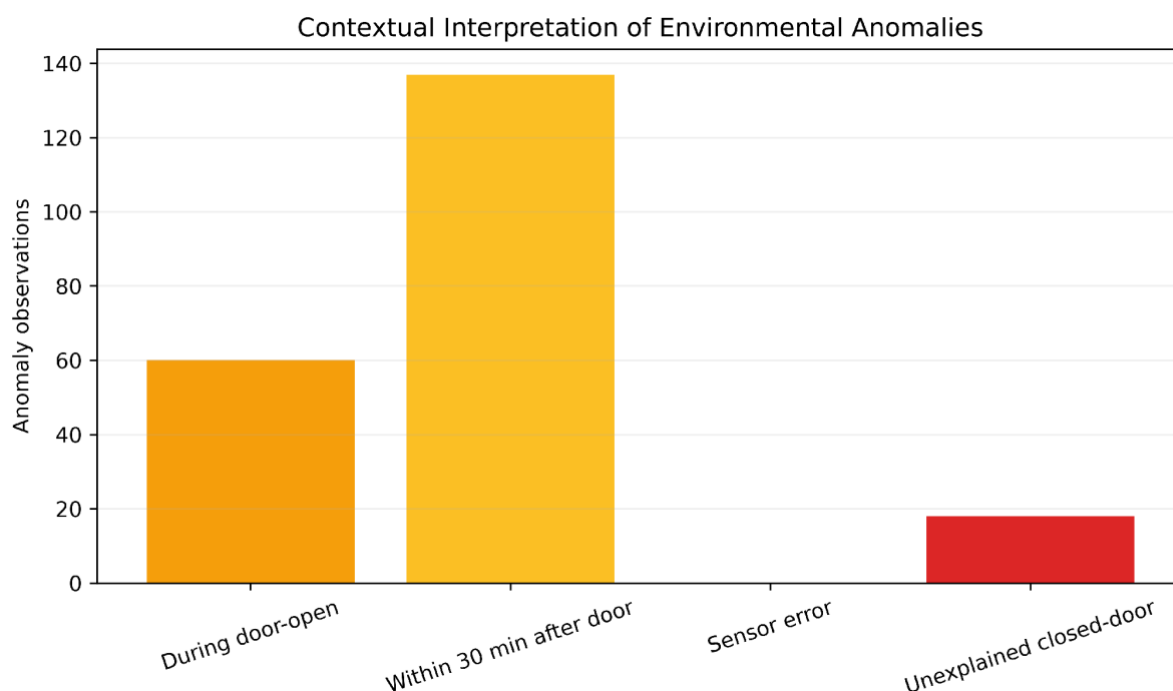


Figure 10 - Contextual classification of refrigerator environmental anomaly observations

The most substantial unexplained closed-door period lasted from 00:38 to 00:52 on 20 August. It contained 15 consecutive flags, with mean humidity reaching 96.15% and mean temperature reaching 8.10°C. The pattern warrants investigation as a possible undocumented handling event, condensation/defrost behaviour, sensor-placement effect or door-state mismatch. It is not evidence of contamination.

5.4 LSTM Results

The 30-observation LSTM achieved recall of 0.9821, precision 0.5884, F1-score 0.7359, PR-AUC 0.8877, ROC-AUC 0.9662 and accuracy 0.8540. It produced 1,976 true positives, 6,318 true negatives, 1,382 false positives and 36 false negatives. The 10-observation model achieved higher precision (0.6470) and accuracy (0.8703), but lower recall (0.8319), F1-score (0.7279), PR-AUC (0.8531) and ROC-AUC (0.9482). The longer history therefore reduced missed occupied observations substantially, although it increased false alarms.

Table 10 - LSTM evaluation on the UCI temporal benchmark

Sequence length	Recall	Precision	F1-score	PR-AUC	ROC-AUC	Accuracy
10	0.8319	0.6470	0.7279	0.8531	0.9482	0.8703
30	0.9821	0.5884	0.7359	0.8877	0.9662	0.8540

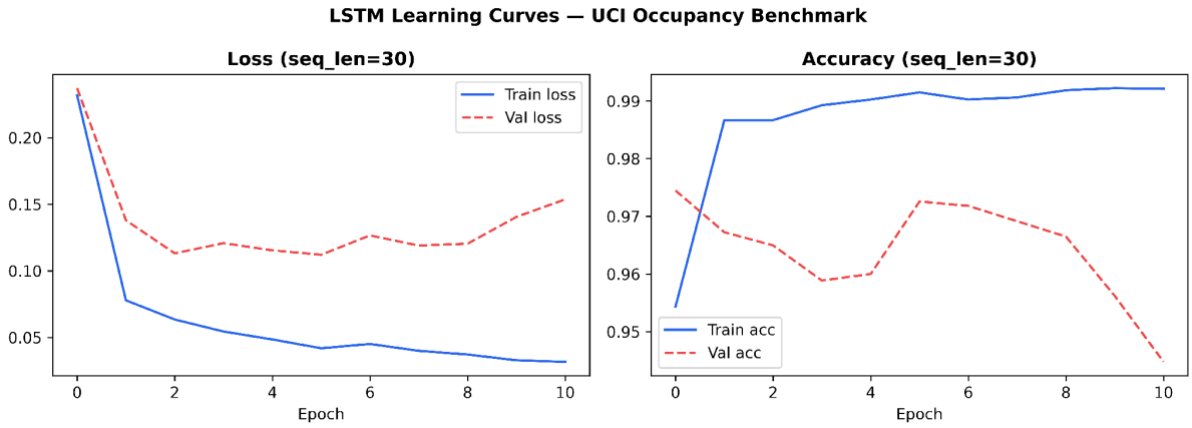


Figure 11 - Training and validation learning curves for the 30-observation LSTM

These results establish temporal classification capability on environmental IoT data. They do not demonstrate food-spoilage prediction because the outcome was occupancy.

5.5 Model Comparison

Table 11 summarizes the completed experiments. Direct ranking is meaningful only within a common dataset and task. On the Fruit classification task, Random Forest substantially exceeded Logistic Regression and the majority baseline. Fruit Isolation Forest achieved similar recall but much lower precision because it was fitted without Bad examples. The LSTM and refrigerator Isolation Forest answer separate temporal and real-world monitoring questions.

Table 11 - Summary of model evaluations

Model	Dataset/task	Recall	Precision	F1-score	PR-AUC	Accuracy
Majority class	Fruit baseline	0.0000	0.0000	0.0000	0.4368	0.5632
Logistic Regression	Fruit supervised	0.9605	0.8617	0.9084	0.9012	0.9154
Random Forest	Fruit supervised	1.0000	0.9981	0.9991	1.0000	0.9992
Isolation Forest	Fruit unsupervised	0.9868	0.6140	0.7570	0.6849	0.7233
Pi reduced RF	Fruit reduced features	0.9737	0.8132	0.8862	0.9099	0.8908
LSTM, 30 observations	UCI temporal benchmark	0.9821	0.5884	0.7359	0.8877	0.8540

Note: The refrigerator Isolation Forest is not included in the supervised performance columns because the refrigerator dataset contains no ground-truth anomaly labels. It flagged 215 observations, of which 91.6% occurred during door opening or within 30 minutes after closure.

McNemar's test was not reported because paired per-observation predictions were not retained for the competing supervised models. Aggregate confusion matrices cannot supply the discordant-pair counts required by the test.

The numerical comparison also reinforces the need to separate predictive performance from deployment usefulness. Random Forest had the strongest labelled accuracy, but it depends on the representativeness of the Fruit labels. The reduced model traded some precision for a feasible sensor set. Isolation Forest supplied a mechanism for previously unseen conditions, but its threshold controlled the alert burden. LSTM captured temporal dependence, while the refrigerator model supplied operationally contextualized outliers without a labelled outcome. No single metric therefore identifies a universally best system component.

5.6 Detection Delay and Environmental Recovery

A valid spoilage-detection delay could not be estimated because the Fruit data contain neither timestamps nor an independently established deterioration onset. The durations of the 17 door-opening episodes are shown in Figure 12. Refrigerator door events provide a different temporal result: physical response and recovery of the monitored environment. The 315 OPEN records formed 17 distinct episodes. Sixteen lasted between one and 23 minutes, while one extended for 257 minutes from 09:33 to 13:50 on 21 August. The long episode raised Sensor 1 by as much as 17.2°C and Sensor 2 by 16.3°C relative to their pre-event baselines.

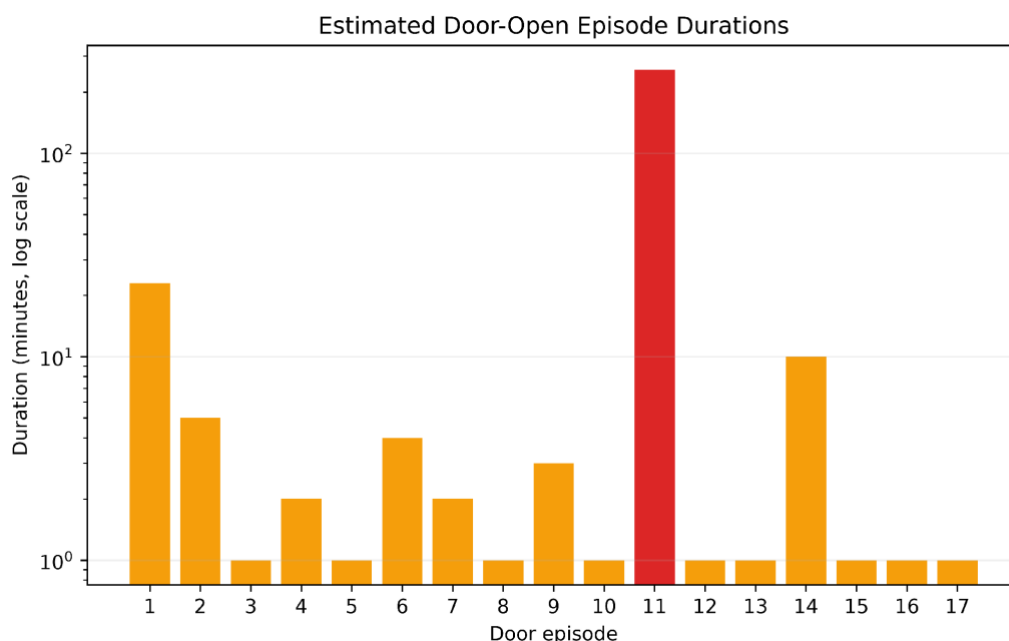


Figure 12 - Estimated durations of the 17-refrigerator door-opening episodes

Only one episode met the deliberately strict dual-sensor recovery rule within the available 120-minute window; it recovered 29 minutes after closure. A representative post-closure temperature response is presented in Figure 13. This result does not mean the refrigerator was unsafe. Recovery required temperature and humidity from both locations to return to robust local baselines for five consecutive observations, and overlapping events shortened some search windows. It should be treated as an analytical operational indicator, not a validated HACCP critical limit.

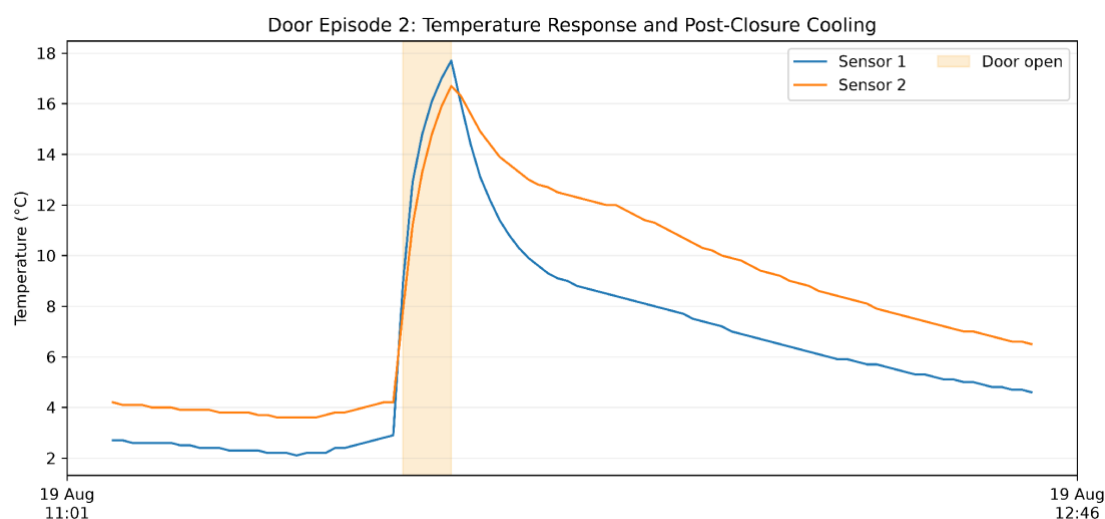


Figure 13 - Temperature response and post-closure cooling for a representative door-opening episode

5.7 Raspberry Pi Prototype Results

The reduced Random Forest achieved recall of 0.9737, precision 0.8132, F1-score 0.8862, PR-AUC 0.9099, ROC-AUC 0.9451 and accuracy 0.8908. Compared with the full model, recall decreased by 2.63 percentage points and precision by 18.49 points. The result supports edge-inference feasibility with a reduced sensor set but does not validate the risk model prospectively on refrigerator data.

The 72-hour refrigerator dataset provides field evidence that the prototype architecture could collect and retain time-stamped measurements from two environmental sensing positions together with door and sensor-health states. It supports environmental monitoring and operational analysis beyond offline simulation. Nevertheless, without the corresponding local Pi CSV, the study cannot determine how many locally generated messages were delivered, delayed, duplicated or lost in transmission. Cloud reliability therefore remains partially, not fully, validated.

Operationally, the refrigerator deployment produced more than a connectivity demonstration. It enabled a distinction between 315 individual OPEN samples and 17 physical door episodes, showed a consistent spatial temperature gradient, and linked most environmental flags to observable door activity. These are important outcomes for a monitoring prototype because they show that the recorded variables can support interpretable event reconstruction rather than merely populate a cloud table.

5.8 SHAP Explainability

Global SHAP analysis ranked light first (mean absolute SHAP 0.1299), followed by temperature-humidity interaction (0.0939), humidity (0.0932), temperature (0.0525), CO₂-light interaction (0.0496), and CO₂ (0.0367). Figure 14 and Figure 15 show the importance, direction and distribution of the SHAP effects. Dependence and summary plots showed that higher light, higher combined temperature-humidity values and the highest humidity values generally shifted the prediction towards Bad. Fruit-category indicators contributed less globally.

Local explanations showed that predictions arose from combinations of measurements rather than a single value. SHAP explains model behaviour, however, and cannot establish a causal mechanism of food deterioration.

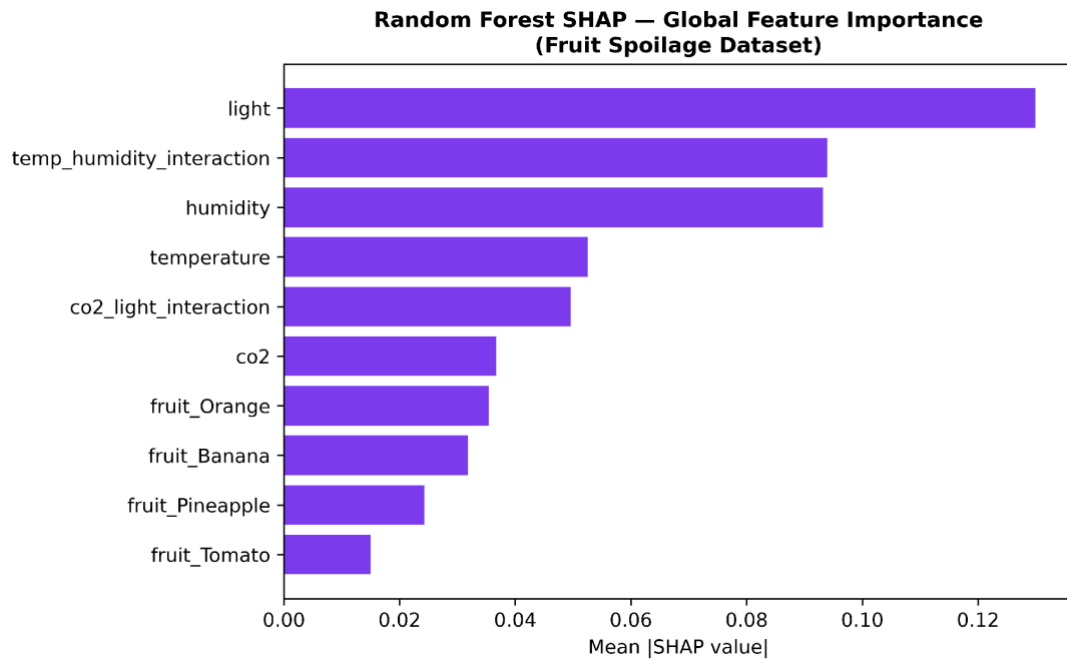


Figure 14 - Global RF importance measured by mean absolute SHAP value

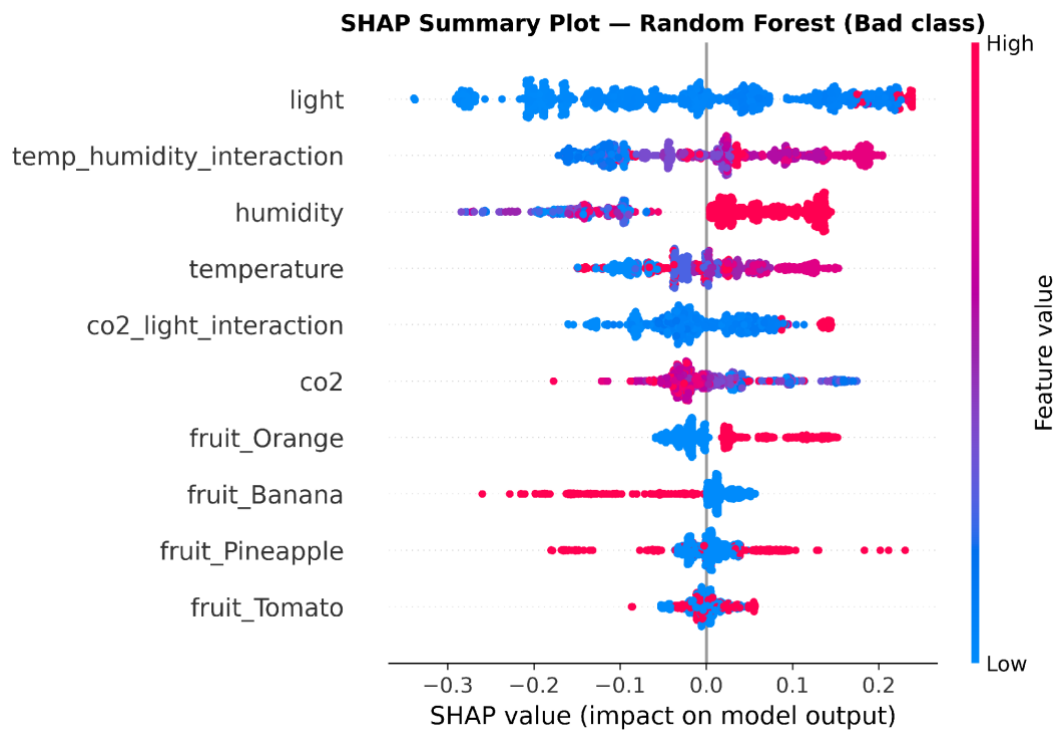


Figure 15 - Direction and distribution of SHAP effects for the RF Bad-class output

Chapter 6 - Discussion

6.1 Interpretation of Findings

The findings support a layered predictive-HACCP proof of concept. Labelled Fruit data showed that environmental variables can distinguish the source dataset's ‘good’ and ‘bad’ classes; UCI data showed that the pipeline can learn from multivariate sequences; and the refrigerator deployment showed how environmental and door-state measurements behave in a real refrigerator. These layers answer different evidential questions and should not be collapsed into one performance claim.

Random Forest produced near-perfect Fruit test performance. Its advantage over Logistic Regression indicates non-linear and interaction-dependent structure, consistent with the importance of temperature-humidity and CO₂-light terms. However, the identical tuned and default validation results, perfect ranking scores and single test error are unusually strong for food operations. They are more likely to reflect a constrained and highly separable public dataset than universal field performance. Duplicate removal prevented direct duplicate leakage, but unavailable batch or acquisition identifiers mean hidden grouping cannot be excluded.

The two Isolation Forest applications clarify the role of unsupervised detection. In the labelled Fruit experiment, high recall required a 48.1% false-positive rate, which would create alert fatigue if deployed without confirmation. In the refrigerator analysis, 91.6% of flags aligned with a door opening or the subsequent 30-minute response. This does not make those observations false alarms; it shows that operational context explains why the environment was unusual. The remaining closed-door periods are investigation priorities, not evidence of contamination. Context-aware anomaly interpretation is therefore more defensible than treating every outlier as a food-safety event.

The refrigerator data also demonstrate why two sensors should not be averaged prematurely. Their trends were strongly correlated, but Sensor 2 was approximately 1.85°C warmer on average. The difference likely reflects placement and local airflow. In practical monitoring, a spatial gradient can be meaningful: a single sensor may underrepresent the warmest location, while uncritical averaging can hide local excursions.

The LSTM result supports the value of temporal context. The 30-observation input increased recall from 0.8319 to 0.9821 and reduced false negatives from 341 to 36, although precision

fell because more unoccupied periods were flagged. This trade-off resembles operational early-warning systems, where longer context can increase sensitivity but also prolong alerts. Because the target was occupancy and light was highly predictive, the result demonstrates temporal modelling capability rather than food-domain accuracy.

6.2 Comparison with Previous Studies

The Random Forest findings are consistent with studies reporting that ensemble methods capture complex relationships in food-quality data (Yi et al., 2024; Yang et al., 2025). The present work similarly found environmental interactions to be influential. Direct numerical comparison is inappropriate, however, because products, targets, sensing technologies and validation designs differ.

The anomaly results align with work showing the value of Isolation Forest for continuously generated IoT data where labelled faults are scarce (Elsaid & Binbusayyis, 2024). The refrigerator evidence adds a practical interpretation: most detected deviations coincided with known door-related disturbances. This supports an event-aware architecture in which door state, maintenance, loading and sensor health accompany the anomaly score. Such context can reduce alert fatigue and make alerts more actionable.

The LSTM benchmark is consistent with reports that recurrent networks can exploit temporal dependencies in freshness-related and environmental sensing (Yuan et al., 2024; Guo & Feng, 2024). The improved recall from the longer window supports the use of sustained patterns rather than isolated readings. The UCI outcome remains weaker domain evidence than a longitudinal food-quality label.

The SHAP results also correspond with explainable food-quality and multivariate sensor studies (Ahmed et al., 2024; Li et al., 2024). Exposing the influence of light, humidity and temperature-humidity interaction makes the classifier auditable. The prominence of light may still reflect dataset acquisition conditions, so explanation should guide scrutiny rather than be treated as causal validation.

An important distinction emerges when SHAP and contextual anomaly interpretation are considered together. SHAP explains why a supervised classifier produced an individual score relative to its learned labels; door context explains the physical circumstances surrounding an unsupervised deviation. Both improve transparency, but neither independently verifies food

condition. A robust industrial system would need to combine statistical explanation, operational metadata and independent product evidence.

Finally, the low-cost edge-cloud architecture resembles Raspberry Pi-based monitoring approaches such as Taylor et al. (2025). The present 72-hour refrigerator record strengthens the implementation evidence by moving beyond offline scripts, but the absence of matched local and cloud logs prevents a quantitative transmission-reliability claim.

6.3 Industrial Applicability

Within a HACCP-based facility, the proposed system is best positioned as decision support. Calibrated sensors could monitor storage zones, equipment and door events; an edge device could calculate features and risk scores; and Azure could provide remote storage, dashboards and audit history. A supervised model could estimate similarity to labelled risk conditions, while an unsupervised layer could identify patterns outside the learned operating envelope.

The refrigerator results suggest that alerts should include context. A message stating only "anomaly" is less useful than one reporting the sensor location, current and recent temperature-humidity trajectory, door status, duration since closure, sensor health and persistence. Staff could then verify equipment, inspect loading or cleaning activity, review time-temperature exposure, check calibration or initiate product assessment under the site's established corrective-action procedure.

The framework must not replace validated critical limits, prerequisite programmes, laboratory testing or professional judgement. Before industrial adoption, product-specific thresholds, tolerable false-alarm rates, sensor calibration schedules, access control, secure credential management and offline buffering would be required. The home refrigerator demonstrates feasibility under a small, controlled deployment, not industrial qualification.

In practice, a staged alert policy would be preferable to a binary automated decision. A short door-related excursion could be logged without escalation; a persistent post-closure deviation could trigger equipment inspection; and an unexplained multi-sensor deviation could request supervisor review or product verification. Such escalation would use the model as an early-warning aid while keeping corrective actions under the validated HACCP plan.

6.4 Generalizability of the Proposed Framework

The trained Fruit classifier has limited direct transferability. It covers four Fruits within narrow environmental ranges and lacks timestamp, facility, batch and microbiological variables. Industrial cold stores introduce packaging, airflow, defrost cycles, loading patterns, seasonal changes, cleaning events and sensor drift. Reusing the model without representative local training data could therefore create both missed hazards and excessive alarms.

The refrigerator deployment improves ecological relevance because it captures genuine time-dependent refrigerator behaviour, including spatial differences and door disturbances. Nevertheless, it covers only one home refrigerator over three days, with no food-condition outcome. It validates sensing and context-aware analysis more strongly than spoilage prediction. The UCI benchmark broadens technical portability across another IoT sensing environment but cannot validate food risk.

What generalizes most credibly is the workflow: preserve raw data; verify sensor quality; retain spatial and operational context; engineer level, trend and volatility features; compare supervised and unsupervised approaches; validate temporally where possible; explain model outputs; and deploy reduced inference at the edge. Each new site would still need local calibration, representative outcome data and prospective evaluation.

6.5 Limitations

The central limitation is the absence of microbiological or independently verified quality outcomes. The Fruit ‘*Bad*’ label supports association-based modelling, while the refrigerator file has no normal/anomaly label at all. Neither can confirm contamination, establish a product-specific critical limit or prove that a detected pattern caused spoilage.

The public Fruit dataset contained 2,880 duplicates and no batch identifiers. Although duplicates were removed before splitting, related observations may remain across partitions. The UCI target is occupancy rather than food safety, and the official validation file is not strictly later than training. Selection of the stronger LSTM sequence using the final period also weakens the independence of the selected result.

The refrigerator deployment lasted approximately three days in one home refrigerator. The 257-minute OPEN period requires experimental documentation before causal interpretation. Recovery criteria and the 30-minute context window are analytical definitions, not validated

HACCP thresholds. The assumed one-minute completeness of the refrigerator file does not measure end-to-end delivery reliability; local Pi records are required for that comparison. Inference latency, power consumption, memory use and behaviour during extended connectivity loss were not quantified. McNemar's test and food-spoilage detection delay were not completed because the retained evidence does not support them. These limitations position the contribution as an explainable AI-enabled IoT monitoring framework and tested proof of concept, rather than a microbiologically validated commercial safety product.

6.6 Future Work

Future work should collect longitudinal data in food-storage or manufacturing environments and link sensor histories to product, batch, location, operational events and independent quality or microbiological measurements. Validation should group observations by batch or collection session and use rolling-origin or forward-chaining evaluation. This would support genuine detection-delay measurement from an independently defined onset to the first sustained alert.

The local Raspberry Pi log should be retained alongside Azure messages with stable message identifiers and timestamps. Record-level matching could then quantify delivery rate, duplication, latency and outage recovery. Door, defrost, loading and cleaning events should be documented explicitly so contextual interpretations can be validated rather than inferred.

Longer deployments should cover multiple refrigerator zones, products, loading levels and seasons. Sensor interchangeability should be evaluated through calibration checks and controlled co-location before spatial differences are interpreted. Repeated door experiments of known duration would allow recovery definitions to be refined and compared with product-specific operating requirements.

Models should be recalibrated using genuine sensor histories. Alert thresholds should be selected with food-safety personnel according to the consequences of false negatives and false positives. Per-observation predictions should be retained for McNemar's test and confidence intervals should be obtained using batch-aware resampling. External validation at another refrigerator or facility should assess transportability.

Finally, explanations should be integrated into the operational interface. Each alert should display risk/anomaly score, contributing variables, sensor health, recent trajectory and relevant HACCP action. These steps would move the framework from a promising proof of concept towards prospective validation.

Chapter 7 - Conclusion and Project Reflection

7.1 Summary of Research

This research developed an explainable AI-enabled IoT framework for monitoring environmental conditions relevant to predictive HACCP. The design combined a labelled public Fruit dataset, a temporal UCI benchmark and a real 72-hour prototype refrigerator deployment. This division allowed the research to examine supervised spoilage-risk classification, unsupervised detection, sequence modelling, edge-compatible inference and real operational environmental behaviour without presenting one dataset as evidence for a task it could not support.

After cleaning and duplicate removal, Random Forest achieved test recall of 1.0000, precision of 0.9981 and F1-score of 0.9991 on the Fruit dataset. Fruit Isolation Forest achieved recall of 0.9868 but precision of 0.6140, demonstrating the false-alarm cost of high-sensitivity unsupervised detection. The 30-observation LSTM achieved recall of 0.9821 and F1-score of 0.7359 on the separate UCI occupancy benchmark. The LSTM results are reported in Table 10, with its learning curves shown in Figure 12 SHAP ranked light, temperature-humidity interaction and humidity as the most influential Random Forest variables.

The refrigerator dataset extended the project beyond offline public data modelling. It contained 4,321 time-stamped observations from two environmental sensing locations, door state, and sensor-health flags. Seventeen door episodes were identified, including one 257-minute period. Context-aware Isolation Forest flagged 215 observations, of which 91.6% occurred during door openings or within 30 minutes after closure. The deployment therefore demonstrates continuous monitoring and operational interpretation, while the absence of outcome labels and a matching local Pi file limits claims about spoilage and cloud transmission reliability.

7.2 Answers to Research Questions

RQ1: Can environmental sensor data be used to identify conditions associated with elevated food-spoilage risk?

Within the labelled Fruit dataset, yes. Environmental measurements and engineered interactions contained strong information about the recorded ‘good’ and ‘bad’ classes, and Random Forest detected all bad test observations. However, the labels were not microbiological outcomes. The conclusion is therefore that sensor data can predict dataset-

defined environmental spoilage-risk classes, not that the model diagnoses contamination or guarantees food safety.

RQ2: How suitable are supervised, unsupervised and temporal AI approaches for different aspects of predictive food-safety monitoring?

Random Forest was the strongest model for the primary labelled Fruit task. It outperformed Logistic Regression and produced only one held-out test error. Isolation Forest was useful when labels were withheld, but high sensitivity created many false alarms in the Fruit experiment. In the real refrigerator data, its value was greatest when combined with door and sensor context. The LSTM demonstrated temporal capability on UCI occupancy but cannot be ranked as a food-spoilage classifier. Thus, the most defensible architecture uses supervised Random Forest where representative labels exist, context-aware anomaly detection where they do not, and temporal models when labelled longitudinal food data become available.

RQ3: How feasible is the deployment of lightweight predictive monitoring on a Raspberry Pi edge device within an IoT environment?

The study partially demonstrated deployment feasibility. Raspberry Pi-based sensing, local display, data logging and Azure telemetry were implemented, while the reduced Random Forest was prepared and evaluated offline. Continuous live model inference was not implemented or operationally validated. The 72-hour refrigerator dataset nevertheless demonstrated that the IoT architecture could continuously collect and retain temperature, humidity and door-state measurements for subsequent analysis. However, record-level Raspberry Pi-to-Azure transmission reliability, inference latency and resilience during network outages were not quantified. Therefore, the findings support the feasibility of the monitoring architecture and offline AI pipeline, but not a fully deployed real-time predictive system. A controlled operational trial integrating continuous edge inference with matched local and cloud records is required before claiming production readiness.

7.3 Research Contributions

The first contribution is an integrated evidence structure for predictive HACCP: labelled food-risk modelling, temporal IoT benchmarking and real-world refrigerator monitoring are connected while their inferential boundaries remain explicit. This avoids describing occupancy as spoilage or an environmental outlier as contamination.

The second contribution is a context-aware interpretation of unsupervised alerts. The refrigerator analysis showed that most unusual observations were physically associated with door operation and recovery. Retaining door state, sensor health and spatial sensor identity makes anomalies more actionable and prevents alarm counts from being mistaken for hazard prevalence.

The third contribution is the combination of performance and transparency. Random Forest supplied high-sensitivity classification, while SHAP exposed the variables and interactions driving predictions. This supports model audit and communication without converting statistical explanation into causal evidence.

The fourth contribution is a reproducible edge-to-cloud proof-of-concept pathway. The project connects cleaning, feature engineering, model export, Raspberry Pi-compatible inference, Azure environmental logging, event analysis and research reporting. Its contribution is not a finished microbiologically validated product; it is a tested framework that shows how AI, IoT monitoring and operational context can be organized to support future predictive-HACCP validation.

Overall, the research demonstrates that the most credible route to predictive HACCP is not to treat every model output as a hazard declaration. It is to combine validated labels where available, unsupervised screening where labels are scarce, temporal and operational context, explainable predictions and auditable sensor records. The project establishes this architecture and identifies the additional microbiological, multi-site and reliability evidence required for responsible progression.

7.4 Project Management Reflection

The original project plan proposed using company data to develop and validate the predictive-HACCP system. However, this data became unavailable, requiring the methodology to be revised. The Multi-Parameter Fruit Spoilage Dataset was therefore selected as the primary food-related dataset, while the UCI Occupancy Dataset was used for temporal modelling. Real-world refrigerator data collected through the Raspberry Pi and Azure prototype provided additional evidence of system operation and environmental behaviour.

Progress generally followed the Gantt chart, although dataset selection, system integration and model refinement required more time than initially planned. Some activities were rescheduled,

but the principal objectives, machine-learning development, IoT implementation, Azure monitoring and dissertation preparation, were completed.

Regular meetings with *Dr. Ann Smith* helped to refine the project scope, strengthen the methodology and ensure that the conclusions remained supported by the available evidence. The feedback was particularly valuable in distinguishing food-spoilage prediction from environmental and anomaly monitoring.

The project demonstrated the importance of contingency planning, accurate documentation and continuous validation. It also improved my skills in machine learning, IoT development, cloud integration, research writing and critical interpretation. Most importantly, I learned to adapt the research plan when data or technical limitations affected the original approach.

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Appendices

Appendix A – Dataset Details and Preparation

This appendix summarizes the three datasets used in the study. Each dataset served a separate analytical purpose, and their results were not treated as directly interchangeable.

Table A.1. Summary of the datasets used in the study

Criterion	Fruit spoilage	UCI occupancy	Prototype refrigerator
Raw observations	10,995	20,560	4,321
Final observations	8,115	20,560	4,321
Structure	Tabular, non-temporal	Time-ordered	Time-ordered, dual-sensor and door
Main variables	Fruit type, temperature, humidity, light, CO ₂	Temperature, humidity, light, CO ₂ , humidity ratio	Two temperature and humidity readings; door state
Target	Good/Bad	Occupied/Unoccupied	None
Primary use	Food-risk classification	Temporal benchmark	Environmental and anomaly monitoring
Main limitation	No timestamps or microbiological ground truth	Not a food-safety dataset	No verified spoilage/anomaly labels

A.1 Multi-Parameter Fruit Spoilage Dataset

The raw dataset contained 10,995 observations and six variables: Fruit type, temperature, relative humidity, light intensity, CO₂ concentration and spoilage class. Column names were standardized, the inconsistent label ‘BAD’ was consolidated with ‘Bad’, and the target was encoded as Good = 0 and Bad = 1.

No missing values were identified. A total of 2,880 exact duplicate rows were removed before partitioning, leaving 8,115 unique observations: 4,569 Good cases (56.3%) and 3,546 Bad cases (43.7%).

Table A.2. Observed ranges in the cleaned Fruit dataset

Variable	Minimum	Maximum
Temperature (°C)	21.00	27.00
Relative humidity (%)	71.00	95.00
Light (lux)	4.22	268.45
CO ₂ (ppm)	20.00	478.00

These ranges were used for plausibility checking rather than as HACCP critical limits. The target was interpreted as Fruit condition or spoilage risk, not confirmed microbiological contamination.

Dataset source: [Multi-parameter Fruit spoilage dataset link](#)

A.2 UCI Occupancy Detection Dataset

The UCI dataset contained 20,560 timestamped observations: 8,143 training records, 2,665 observations in the first test file and 9,752 in the second. Timestamps were parsed, records were sorted within their original files, numeric columns were converted explicitly and the occupancy target was stored as an integer. No missing values or exact duplicates were identified. The dataset was used only as an LSTM temporal benchmark; occupancy was not interpreted as food spoilage.

Dataset source: [UCI occupancy detection dataset link](#)

A.3 Prototype Refrigerator Dataset

The prototype dataset contained 4,321 timestamped observations collected over 72 hours from two temperature-humidity sensors and a door-state sensor. The median sampling interval was 60 seconds, and the maximum was 73 seconds. Sensor 1 contained three error records and Sensor 2 contained none. The dataset had no ground-truth spoilage or anomaly label; findings were therefore interpreted as environmental and operational observations.

Appendix B – Preprocessing and Feature Engineering

Table B.1. Preprocessing performed for each dataset

Dataset	Preprocessing
Fruit spoilage	Standardized names and labels; removed duplicates; encoded target; one-hot encoded Fruit type.
UCI occupancy	Parsed timestamps; retained original file boundaries; converted numeric variables; checked missingness and duplicates; fitted scaling on training data only.
Refrigerator	Parsed and sorted timestamps; checked schema and intervals; retained error flags; normalized door state; limited interpolation to feature calculation.

All raw source files were retained unchanged. Processed and engineered versions were written separately to preserve traceability.

B.1 Engineered Features

- Fruit: temperature $\times$ humidity and CO₂ $\times$ light interactions; nominal one-hot encoding of Fruit type.
- UCI: lags of 1, 5 and 10 observations; rolling statistics over 10 observations; first differences; hour and day features. The final LSTM used 16 focused inputs and sequences of 10 and 30 observations.
- Refrigerator: separate and mean sensor values; sensor differences; rates of change; 1-, 5- and 15-minute deltas; rolling statistics over 5, 15 and 30 minutes; door-event and recovery context.
- Reduced edge model: temperature, humidity, their interaction, rolling means and standard deviations, first differences, door-open count and hour of day.

Door state and sensor-error indicators were excluded from Azure Isolation Forest fitting and used afterwards for contextual interpretation. The reduced edge experiment assessed software and feature feasibility rather than a validated real-time refrigerator-risk model.

Appendix C – Model Configurations

A random seed of 42 was used for conventional machine-learning experiments.

Table C.1. Final model configurations

Model	Dataset	Configuration
Random Forest	Fruit	100 trees; unrestricted depth; sqrt feature sampling; minimum leaf 1; balanced weights.
Fruit Isolation Forest	Fruit	Trained on Good observations only; candidate contamination 0.05–0.48; selected using validation recall.

Model	Dataset	Configuration
LSTM	UCI	64- and 32-unit recurrent layers; dropout 0.30; batch normalization ; 32-unit ReLU; sigmoid output.
Azure Isolation Forest	Refrigerator	300 trees; contamination 0.05; standardized features; seed 42.
Reduced Random Forest	Reduced Fruit features	200 trees; maximum depth 15; minimum leaf 2.

C.1 Training and Validation

Random Forest tuning considered 100, 200 and 300 trees; unrestricted, 10 and 20 maximum depths; leaf sizes of 1, 2 and 4; square-root and $\log_2$ feature sampling; and balanced-weight alternatives. Recall for the Bad class was prioritized.

The LSTM used Adam optimization , binary cross-entropy loss and class weighting. Sequence lengths of 10 and 30 were trained for up to 30 epochs with batch size 128. Early stopping restored the best weights, and learning-rate reduction was applied on a plateau. A 0.50 threshold converted probabilities into occupancy predictions.

Azure Isolation Forest sensitivity was examined at contamination values 0.01, 0.025, 0.05, 0.075 and 0.10. Contamination was treated as an operating parameter rather than hazard prevalence.

Appendix D – Raspberry Pi Prototype

The prototype was constructed around a Raspberry Pi 4 with two DHT22 temperature-humidity sensors, a magnetic reed switch, LCD, breadboard, jumper wires, development LED, Wi-Fi connection, uninterruptible power supply and temporary enclosure. One DHT22 sensor was placed near the refrigerator door and the other towards the back.

Table D.1. Prototype connections

Component	Purpose	Connection
DHT22 Sensor 1	Door-area temperature and humidity	GPIO 17
DHT22 Sensor 2	Rear-area temperature and humidity	GPIO 24
Reed switch	Door OPEN/CLOSED state	GPIO 5
LCD	Local display	GPIO 2 and 3
LED	Door opening indicator (door status)	GPIO

D.1 Deployment

The prototype operated in one domestic refrigerator for 72 hours at an intended one-minute interval. The Raspberry Pi timestamped each observation, stored it locally, updated the LCD and transmitted telemetry through Wi-Fi to Azure IoT Hub. Seventeen door-opening episodes were identified, including one exceptional episode lasting approximately 257 minutes. This was a proof-of-concept deployment rather than an industrial HACCP trial.

Appendix E – Source Code and Reproducibility

Project title: AI-Enabled Predictive HACCP Using IoT-Based Environmental Monitoring

Repository: [Source code in GitHub repository](#)

Repository structure:

ai-enabled-predictive-haccp-iot-monitoring

```
|— README.md
|— requirements.txt
|— .gitignore
|— data/
|   |— raw/
|   |   |— Fruit_spoilage/
|   |   |   |— uci_occupancy/
|   |   |— processed/
|   |   |   |— Fruit_spoilage_clean.cs
|   |   |   |— Fruit_spoilage_engineered.csv
|   |   |   |— uci_occupancy/
|— models/
|   |— Fruit_random_forest.joblib
|   |— Fruit_isolation_forest.joblib
|   |— pi_reduced_rf.joblib
|   |— uci_lstm_best.keras
|   |— uci_lstm_scaler.joblib
|— machine_learning/
|   |— __init__.py
|   |— data/
|   |   |— load_data.py
|   |   |— clean_Fruit.py
|   |   |   |— clean_occupancy.py
|   |— features/
|   |   |— feature_engineering.py
|   |— models/
|   |   |— train_baselines.py
|   |   |— train_random_forest.py
```

```

| | | └─ train_isolation_forest.py
| | └─ train_lstm.py
| └─ evaluation/
| | └─ comparative_evaluation.py
| | └─ shap_analysis.py
| └─ deployment/
|   └─ train_pi_model.py
└─ raspberry_pi/
  └─ README.md
  └─ requirements.txt
  └─ config.example.json
  └─ sensor_reader.py
  └─ inference.py
  └─ azure_sender.py
└─ notebooks/
  └─ 01_exploratory_data_analysis.ipynb
└─ reports/
  └─ figures/
    └─ eda/
      └─ Fruit_spoilage/
        └─ shap/
          └─ uci_occupancy/
            └─ tables/
              └─ experiment_logs/

```

E.1 Reproduction Summary

1. Download the datasets identified in Dataset/README.md.
2. Install the packages listed in requirements.txt.
3. Run preprocessing and feature engineering.
4. Train the Random Forest, Isolation Forest and LSTM models.
5. Run evaluation and SHAP analysis.
6. Export the reduced Random Forest and preprocessing objects as a *joblib* artefact.
7. Follow raspberry_pi_prototype/README.md to configure the prototype.

Appendix F – Data Management, Ethics and Safety

The study used two public datasets and one environmental dataset collected from the author's refrigerator prototype. No human participants, personal information or biological samples were involved. The prototype measured temperature, humidity and refrigerator door state only.

Raw datasets were retained unchanged and processed versions were stored separately. Azure credentials were separated from analytical data and excluded from the public repository. The low-voltage prototype was treated as a research proof of concept and was not used to determine whether food was safe for consumption.

Ethics status: signed Ethical Review Form submitted as a separate file.

Risk-assessment reference: this file has been included in Terms of reference file of section 4.2.

Appendix G – Project Management Evidence

The original plan proposed using company data. When those data became unavailable, the methodology was revised to use a labelled Fruit-spoilage dataset, the UCI Occupancy Detection Dataset and a 72-hour Raspberry Pi–Azure refrigerator dataset.

- Literature review and research-gap identification
- Dataset selection and validation
- Data cleaning and feature engineering
- Random Forest, Isolation Forest and LSTM development
- SHAP-based interpretation
- Raspberry Pi prototype construction
- Azure IoT telemetry collection
- Refrigerator deployment and contextual anomaly analysis
- Dissertation preparation and repository organization
- The Terms of References (ToR) is attached as a separate file as a supplementary file.

Phase	Activity	Deliverable	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Planning	Proposal Submission	Due 3 Mar								
Literature	Literature Review									
Ethics & ToR	Ethics Form	Due 15 May								
Ethics & ToR	Terms of Reference	Due 22 May								
IoT Dev	Hardware Setup & Azure Config									
Data	Data Collection & Preprocessing									
ML Models	Random Forest									
ML Models	LSTM									
ML Models	Isolation Forest									
Evaluation	Model Evaluation & SHAP									
Writing	Dissertation Writing									
Submission	Final Submission & Presentation	Due 1 Sep								

Figure G.1: Final Gantt chart showing planned and completed activities